Supplementary Material

Standardised Comparison of Convolutional and Transformer Backbones for Brain Tumour MRI Classification under Perceptual-Hash Deduplication

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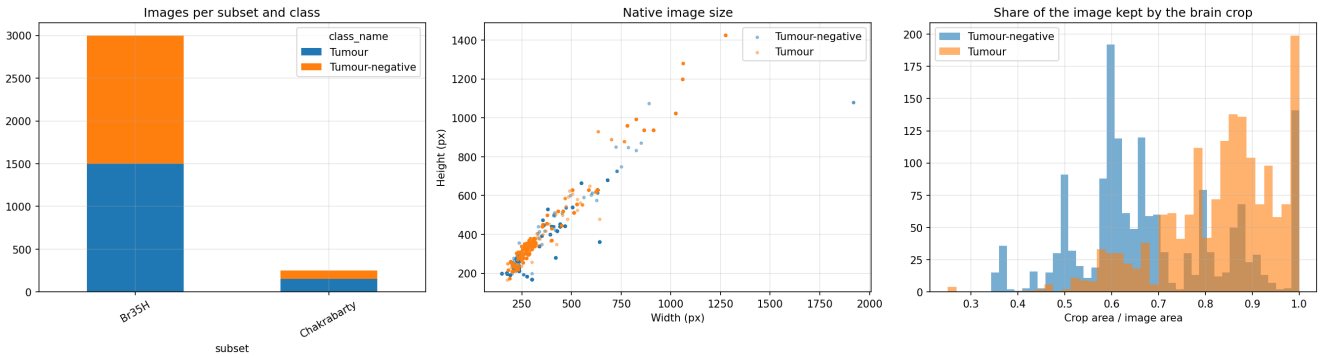
This document accompanies the main manuscript. Every value is taken from the outputs of the two notebooks released with the code repository (`Materials/Notebooks/binary-detection` and `Materials/Notebooks/classification`); the source file is named in each caption. Unless stated otherwise, binary detection is evaluated on 171 held-out test images and subtype classification on 45.

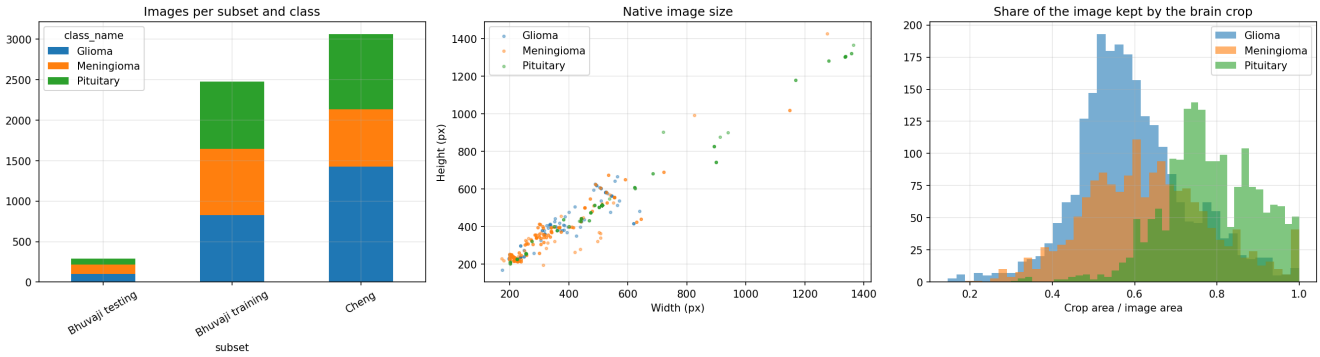
S1. Data, deduplication and splitting

Table S1: Images kept by the label-aware perceptual-hash pass at each Hamming tolerance (`deduplication_threshold_sweep.csv`). The study uses tolerance 5. The multiclass count at tolerance 5 is before the label-noise exclusion, which brings the pool to 4,730.

Tolerance	0, 1	2, 3	4, 5	6, 7	8, 9	10, 11	12
Binary, tumour-negative	860	583	455	395	341	310	285
Binary, tumour	1,217	912	730	607	541	494	424
Binary, total	**2,077**	**1,495**	**1,185**	**1,002**	**882**	**804**	**709**
Multiclass, glioma	2,303	2,221	2,010	1,653	1,277	919	627
Multiclass, meningioma	1,518	1,453	1,316	1,130	935	704	506
Multiclass, pituitary	1,751	1,663	1,408	1,145	886	663	478
Multiclass, total	**5,572**	**5,337**	**4,734**	**3,928**	**3,098**	**2,286**	**1,611**

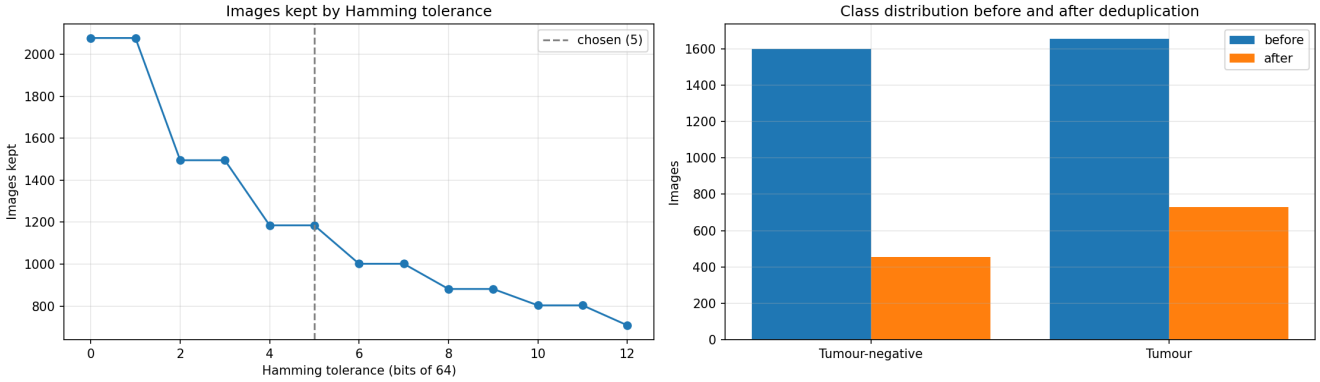


(a) Binary detection.

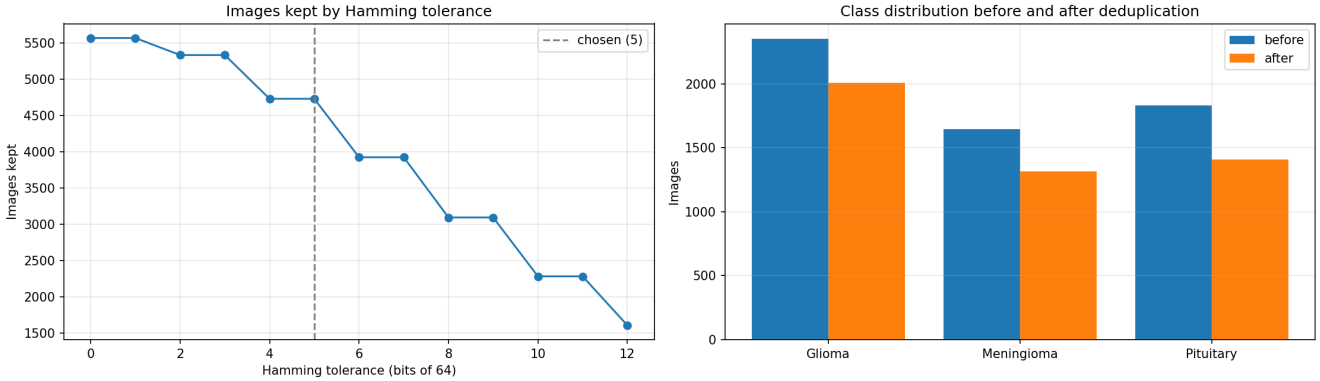


(b) Subtype classification.

Figure S1: Dataset overview: images per source and class, and native image sizes, before deduplication.



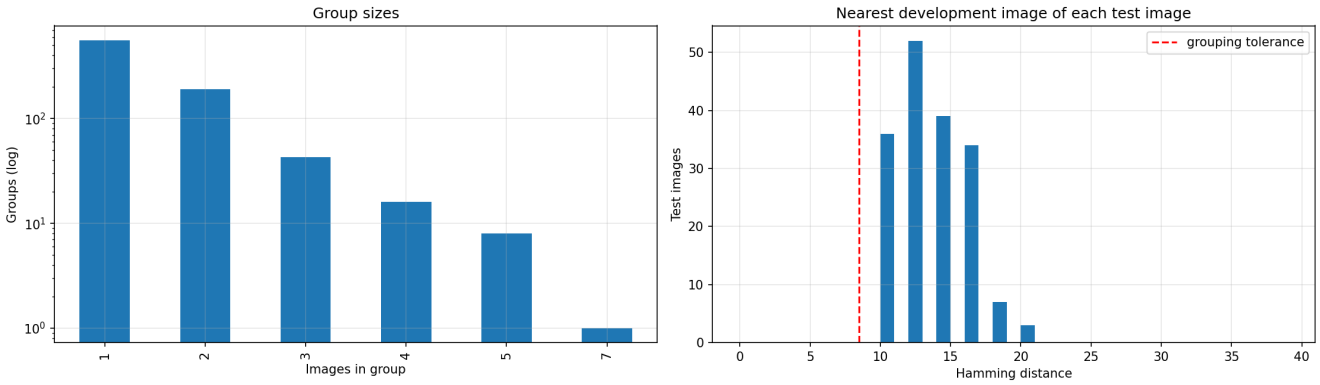
(a) Binary detection.



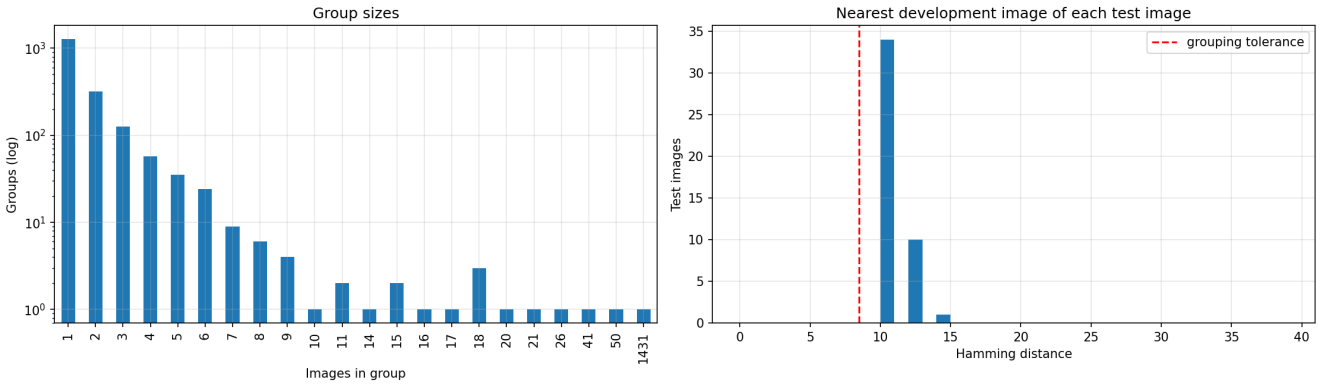
(b) Subtype classification.

Figure S2: Images kept at each Hamming tolerance, and the class distribution before and after the pass at tolerance 5.

Figure S3: Multiclass cross-label pairs within Hamming distance 5, label-noise pairs first. All 31 pairs are glioma against meningioma; the 8 pairs at NCC ≥ 0.98 lie inside the Bhuvaji collection and both copies were excluded, and the 23 others are hash false positives and were kept. The binary pool has no cross-label pair.

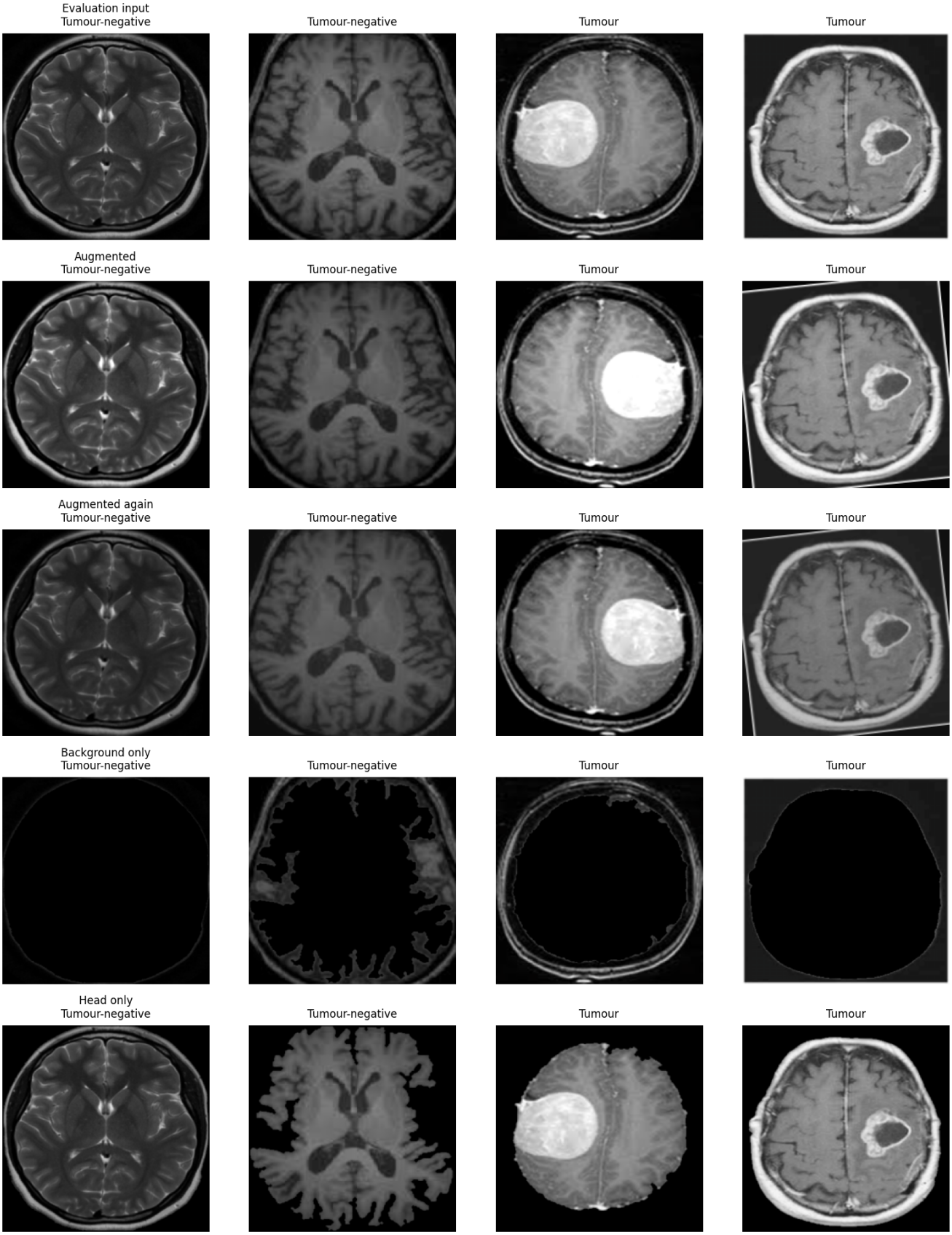


(a) Binary detection: 821 groups, the largest of 7 images.

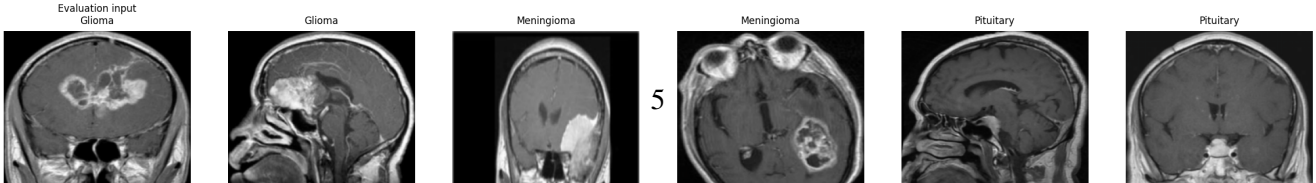


(b) Subtype classification: 1,867 groups, the largest of 1,431 images.

Figure S4: Group sizes and the distance from each test image to its nearest development image. No test image has a development image within Hamming distance 5 or 8 or at NCC ≥ 0.98 (`leakage_audit.csv`).



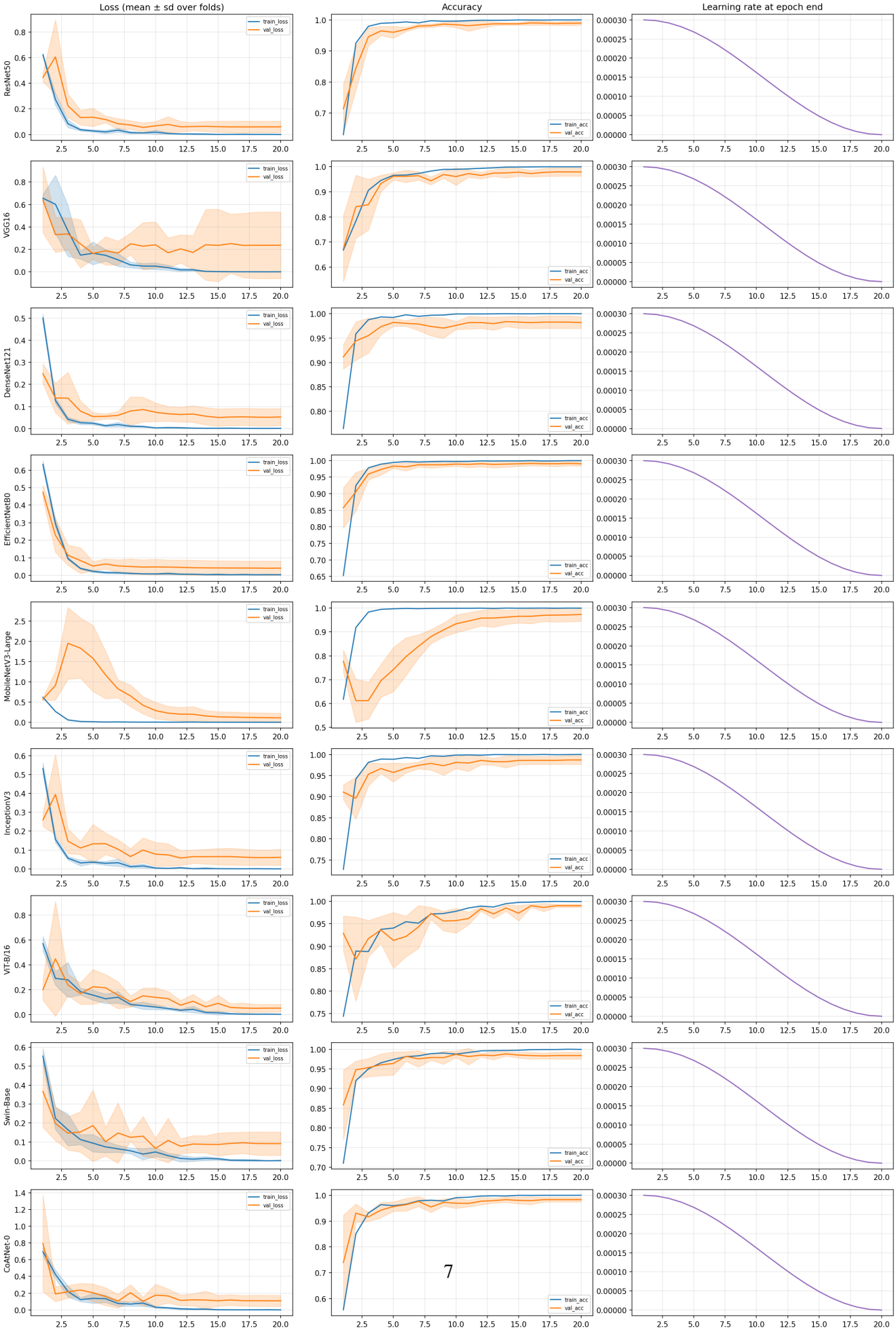
(a) Binary detection.



S2. Training

Table S2: Binary detection: every full-image run (`run_metrics.csv`). Every run took 260 optimiser steps. T is the temperature fitted on the run's validation fold and Thr. the Youden threshold chosen there; accuracy and log-loss are on the 171 test images after temperature scaling. In fold 4 the temperature of six models lies at or near the lower search bound $e^{-3} \approx 0.0498$.

Model	Fold	Best epoch	T	Thr.	Test accuracy	Test log-loss	Train images
CoAtNet-0	0	9	1.1767	0.739	0.9883	0.0491	830
	1	7	1.4761	0.529	0.9825	0.0534	803
	2	12	1.7756	0.633	0.9474	0.1317	807
	3	14	2.1222	0.316	0.9883	0.0333	804
	4	7	0.0604	1.000	0.9708	0.8079	812
DenseNet121	0	6	0.8003	0.724	0.9766	0.0847	830
	1	11	1.0106	0.657	0.9649	0.0751	803
	2	12	1.3466	0.885	0.9649	0.1019	807
	3	14	1.1351	0.950	0.9825	0.0696	804
	4	15	0.5882	0.927	0.9708	0.2513	812
EfficientNetB0	0	5	1.2527	0.621	0.9708	0.0875	830
	1	14	1.2311	0.962	0.9766	0.0378	803
	2	13	1.1739	0.625	0.9825	0.0471	807
	3	11	0.9215	0.985	0.9825	0.0755	804
	4	19	0.0498	1.000	0.9766	0.5431	812
InceptionV3	0	17	1.6119	0.895	0.9766	0.0543	830
	1	8	1.4656	0.045	0.9766	0.0603	803
	2	10	0.9470	0.861	0.9766	0.1005	807
	3	5	1.1069	0.829	0.9825	0.0372	804
	4	12	0.0682	1.000	0.9825	0.3662	812
MobileNetV3-Large	0	20	2.7638	0.121	0.9298	0.1551	830
	1	20	1.8160	0.583	0.9766	0.0638	803
	2	14	1.6694	0.410	0.9883	0.0448	807
	3	20	1.7832	0.249	0.9591	0.1161	804
	4	13	0.0508	1.000	0.9649	0.7224	812
ResNet50	0	9	1.7422	0.819	0.9474	0.1451	830
	1	8	1.7905	0.314	0.9766	0.0837	803
	2	7	1.0362	0.853	0.9649	0.0787	807
	3	17	1.2096	0.759	0.9708	0.1416	804
	4	12	0.0498	1.000	0.9825	0.4848	812
Swin-Base	0	14	1.7903	0.617	0.9708	0.0826	830
	1	11	1.7026	0.550	0.9708	0.1311	803
	2	6	1.5159	0.683	0.9591	0.1221	807
	3	5	0.8935	0.261	0.9883	0.0536	804
	4	10	0.0498	1.000	1.0000	0.0000	812
VGG16	0	5	1.7926	0.462	0.9532	0.1252	830
	1	13	2.1367	0.216	0.9649	0.0912	803
	2	18	1.5436	0.775	0.9649	0.2265	807
	3	4	1.4973	0.196	0.9474	0.1276	804
	4	6	1.8620	0.495	0.9649	0.1407	812
ViT-B/16	0	17	1.4451	0.767	0.9825	0.0566	830
	1	12	1.5874	0.483	0.9649	0.1114	803
	2	17	1.3066	0.824	0.9883	0.0526	807
	3	8	0.8068	0.559	0.9708	0.0711	804
	4	16	0.9993	0.836	0.9708	0.1007	812


Loss (mean ± sd over folds)
Accuracy
Learning rate at epoch end
ResNet50
VGG16
DenseNet121
EfficientNetB0
MobileNetV3-Large
InceptionV3
ViT-B/16
Swin-Base
CoAtNet-0
train_loss
val_loss
train_acc
val_acc

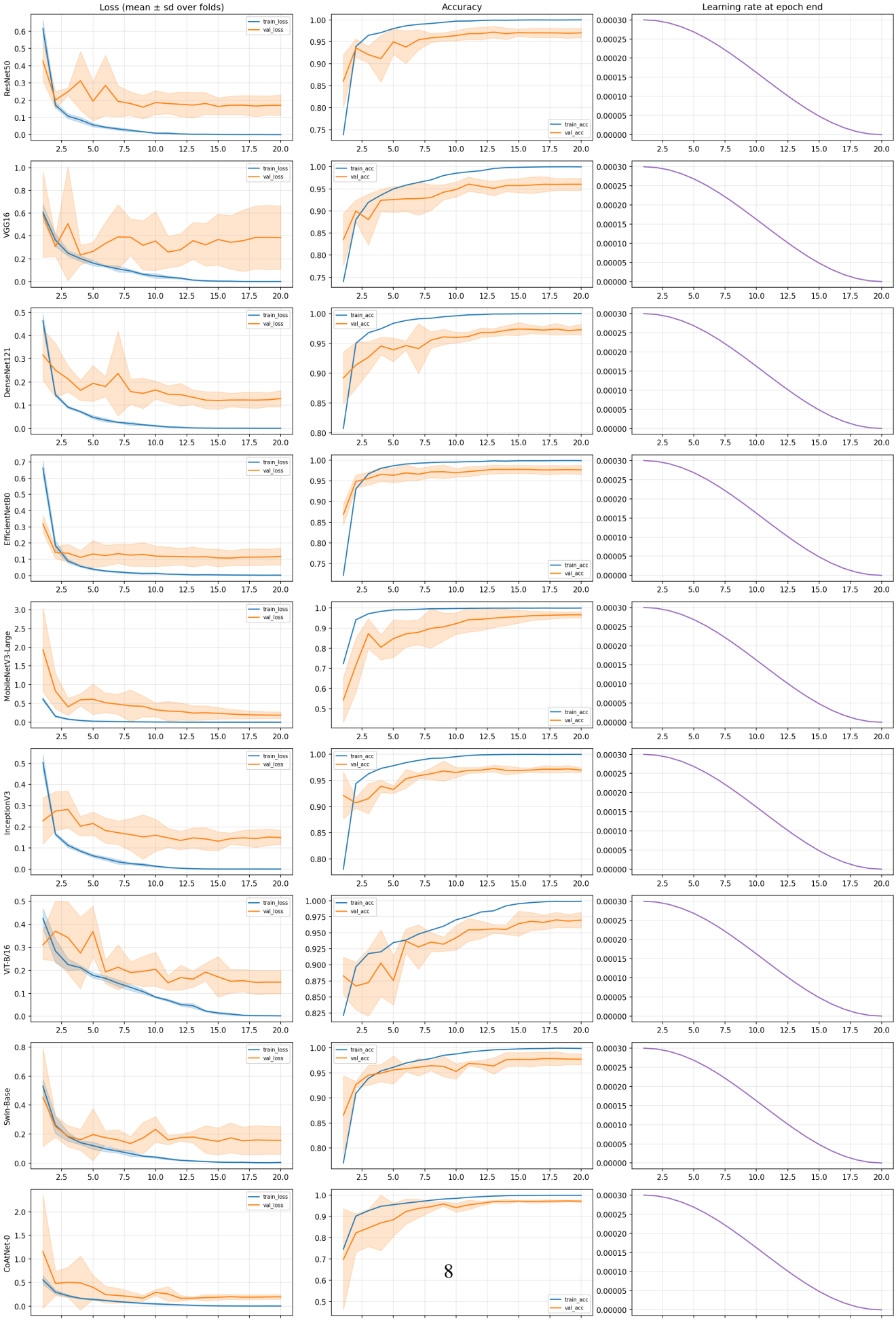

Loss (mean ± sd over folds)
Accuracy
Learning rate at epoch end
ResNet50
VGG16
DenseNet121
EfficientNetB0
MobileNetV3-Large
InceptionV3
ViT-B/16
Swin-Base
CoAtNet-0
train_loss
val_loss
train_acc
val_acc

Table S3: Subtype classification: every full-image run (`run_metrics.csv`). Optimiser steps depend on the fold because the group-aware folds differ in size; within a fold they are equal for every model. In fold 0, whose validation fold has 154 images, the temperature of EfficientNetB0, InceptionV3 and Swin-Base lies at or near the lower search bound.

Model	Fold	Steps	Best epoch	T	Test accuracy	Test log-loss	Train / val images
CoAtNet-0	0	1,420	7	0.8899	1.0000	0.0209	4,531 / 154
	1	1,240	9	1.3042	1.0000	0.0034	3,943 / 742
	2	1,260	5	0.8503	0.9778	0.0720	3,988 / 697
	3	1,200	6	1.2193	0.9778	0.0395	3,809 / 876
	4	780	15	2.3073	0.9778	0.0236	2,469 / 2,216
DenseNet121	0	1,420	7	1.1754	1.0000	0.0040	4,531 / 154
	1	1,240	15	1.9404	1.0000	0.0061	3,943 / 742
	2	1,260	9	1.5745	1.0000	0.0186	3,988 / 697
	3	1,200	4	1.4594	0.9778	0.1067	3,809 / 876
	4	780	10	1.5314	0.9778	0.0355	2,469 / 2,216
EfficientNetB0	0	1,420	9	0.0498	0.9778	0.6140	4,531 / 154
	1	1,240	4	1.5348	0.9111	0.1585	3,943 / 742
	2	1,260	15	1.8186	1.0000	0.0024	3,988 / 697
	3	1,200	3	1.3621	1.0000	0.0181	3,809 / 876
	4	780	16	1.7429	0.9778	0.0330	2,469 / 2,216
InceptionV3	0	1,420	9	0.0515	1.0000	0.0000	4,531 / 154
	1	1,240	7	1.5646	1.0000	0.0050	3,943 / 742
	2	1,260	10	1.6805	1.0000	0.0040	3,988 / 697
	3	1,200	15	2.2675	1.0000	0.0101	3,809 / 876
	4	780	11	1.7383	0.9778	0.0503	2,469 / 2,216
MobileNetV3-Large	0	1,420	12	1.0519	0.9778	0.1369	4,531 / 154
	1	1,240	20	3.1578	0.9778	0.0712	3,943 / 742
	2	1,260	18	2.6818	0.9778	0.0685	3,988 / 697
	3	1,200	8	2.4891	0.9778	0.0327	3,809 / 876
	4	780	20	3.0036	0.9778	0.0689	2,469 / 2,216
ResNet50	0	1,420	5	0.9164	1.0000	0.0026	4,531 / 154
	1	1,240	9	2.0543	0.9778	0.0395	3,943 / 742
	2	1,260	5	1.5465	0.9778	0.0672	3,988 / 697
	3	1,200	15	2.6184	1.0000	0.0176	3,809 / 876
	4	780	8	1.6629	0.9778	0.0957	2,469 / 2,216
Swin-Base	0	1,420	15	0.0498	1.0000	0.0000	4,531 / 154
	1	1,240	8	1.6375	1.0000	0.0112	3,943 / 742
	2	1,260	9	1.9777	1.0000	0.0076	3,988 / 697
	3	1,200	5	1.7771	0.9778	0.0397	3,809 / 876
	4	780	18	2.2403	1.0000	0.0047	2,469 / 2,216
VGG16	0	1,420	11	1.2077	0.9778	0.1367	4,531 / 154
	1	1,240	7	1.7136	1.0000	0.0175	3,943 / 742
	2	1,260	2	1.6416	0.9333	0.2497	3,988 / 697
	3	1,200	4	2.0744	0.9778	0.0520	3,809 / 876
	4	780	13	2.2774	0.9778	0.0970	2,469 / 2,216
ViT-B/16	0	1,420	15	1.3910	1.0000	0.0020	4,531 / 154
	1	1,240	13	1.7649	1.0000	0.0238	3,943 / 742
	2	1,260	13	1.2801	0.9778	0.0420	3,988 / 697
	3	1,200	11	1.5805	1.0000	0.0129	3,809 / 876
	4	780	11	1.4512	0.9778	0.1176	2,469 / 2,216

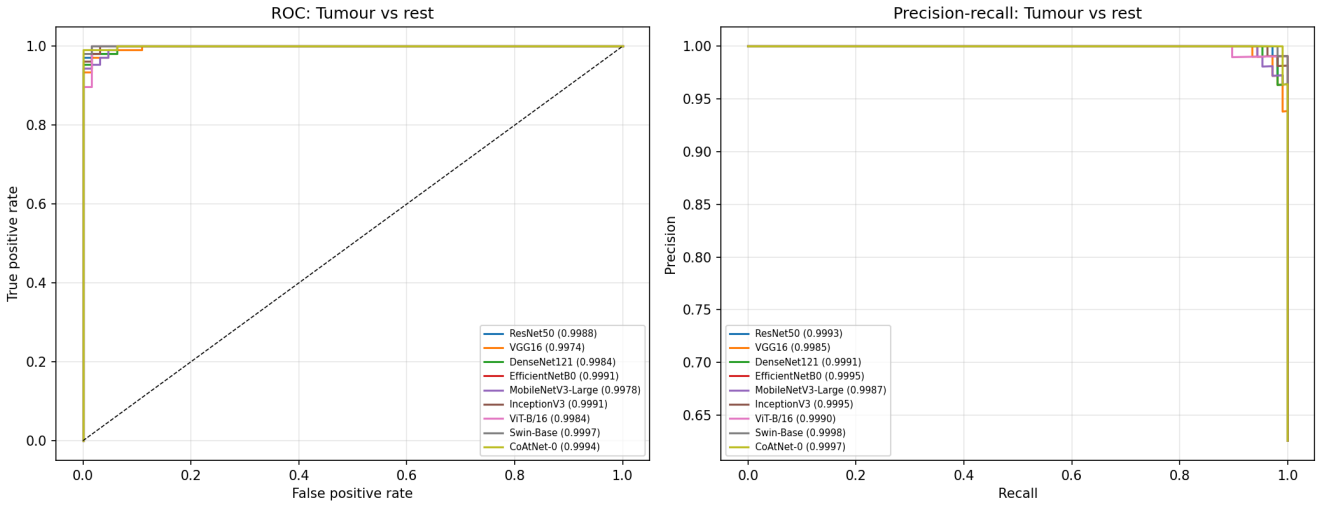
S3. Test results

Table S4: Per-class test metrics of the fold ensembles (`test_per_class.csv`). Binary support: 64 tumour-negative, 107 tumour. Multiclass support: 1 glioma, 13 meningioma, 31 pituitary.

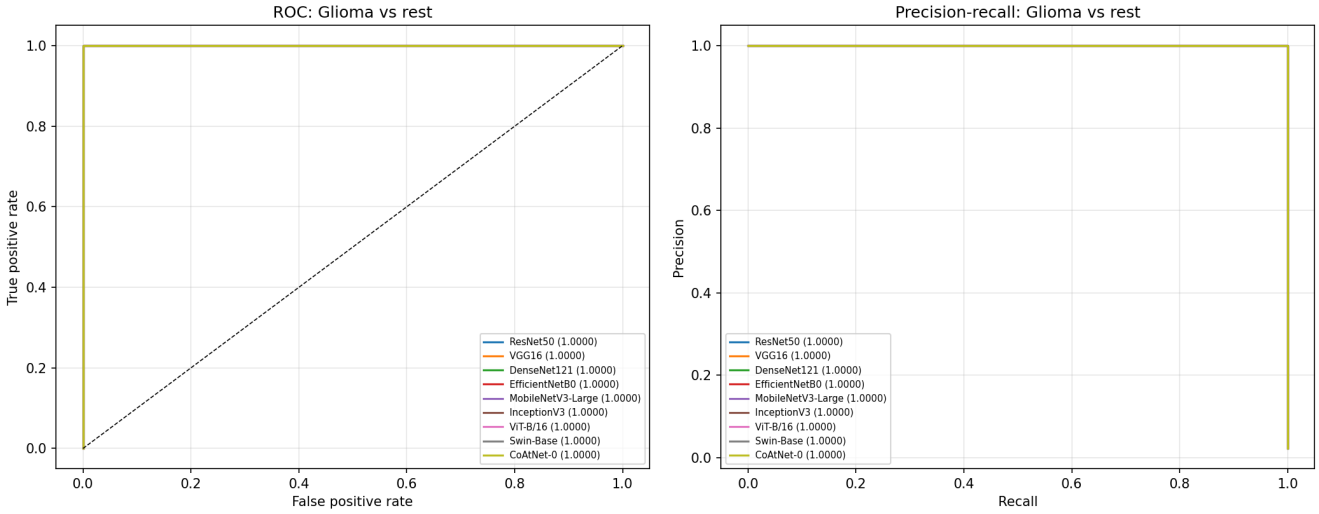
Model	Class	TP	FP	FN	Precision	Sensitivity	Specificity
ResNet50	Tumour-negative	63	2	1	0.9692	0.9844	0.9813
	Tumour	105	1	2	0.9906	0.9813	0.9844
VGG16	Tumour-negative	62	3	2	0.9538	0.9688	0.9720
	Tumour	104	2	3	0.9811	0.9720	0.9688
DenseNet121	Tumour-negative	60	1	4	0.9836	0.9375	0.9907
	Tumour	106	4	1	0.9636	0.9907	0.9375
EfficientNetB0	Tumour-negative	63	0	1	1.0000	0.9844	1.0000
	Tumour	107	1	0	0.9907	1.0000	0.9844
MobileNetV3-Large	Tumour-negative	62	3	2	0.9538	0.9688	0.9720
	Tumour	104	2	3	0.9811	0.9720	0.9688
InceptionV3	Tumour-negative	62	1	2	0.9841	0.9688	0.9907
	Tumour	106	2	1	0.9815	0.9907	0.9688
ViT-B/16	Tumour-negative	63	0	1	1.0000	0.9844	1.0000
	Tumour	107	1	0	0.9907	1.0000	0.9844
Swin-Base	Tumour-negative	64	2	0	0.9697	1.0000	0.9813
	Tumour	105	0	2	1.0000	0.9813	1.0000
CoAtNet-0	Tumour-negative	64	1	0	0.9846	1.0000	0.9907
	Tumour	106	0	1	1.0000	0.9907	1.0000
ResNet50, DenseNet121, InceptionV3, ViT-B/16, Swin-Base, CoAtNet-0	Glioma	1	0	0	1.0000	1.0000	1.0000
	Meningioma	13	0	0	1.0000	1.0000	1.0000
	Pituitary	31	0	0	1.0000	1.0000	1.0000
VGG16, EfficientNetB0, MobileNetV3-Large	Glioma	1	1	0	0.5000	1.0000	0.9773
	Meningioma	12	0	1	1.0000	0.9231	1.0000
	Pituitary	31	0	0	1.0000	1.0000	1.0000

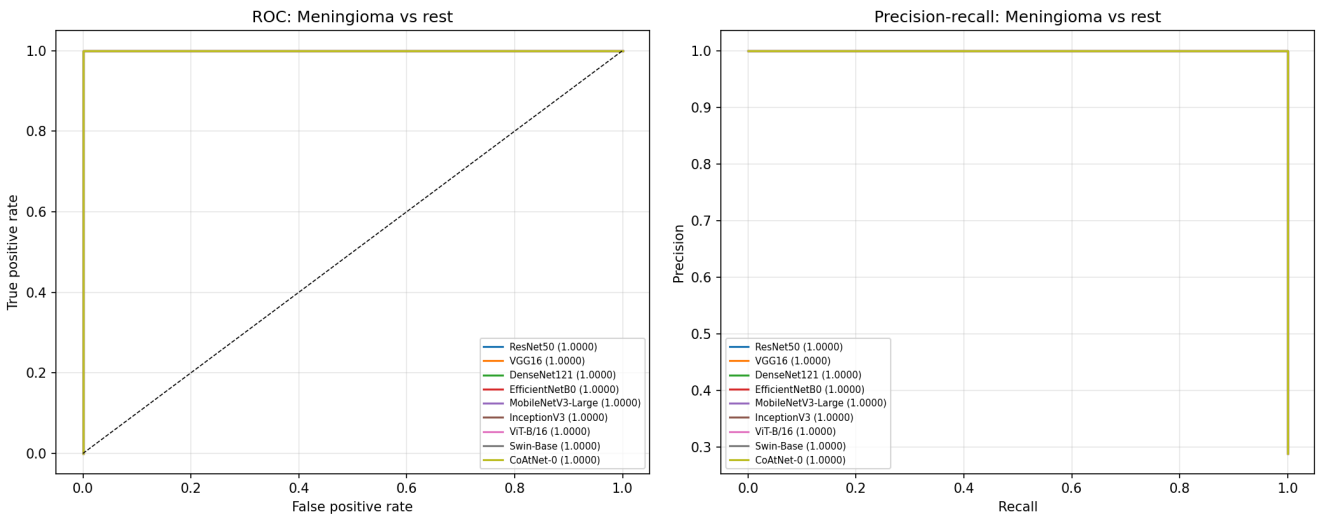
Table S5: Fold-ensemble test accuracy by the Hamming distance from each test image to its nearest development image (`accuracy_by_nearest_dev_distance.csv`). The multiclass test set has no image at distance 17 to 20.

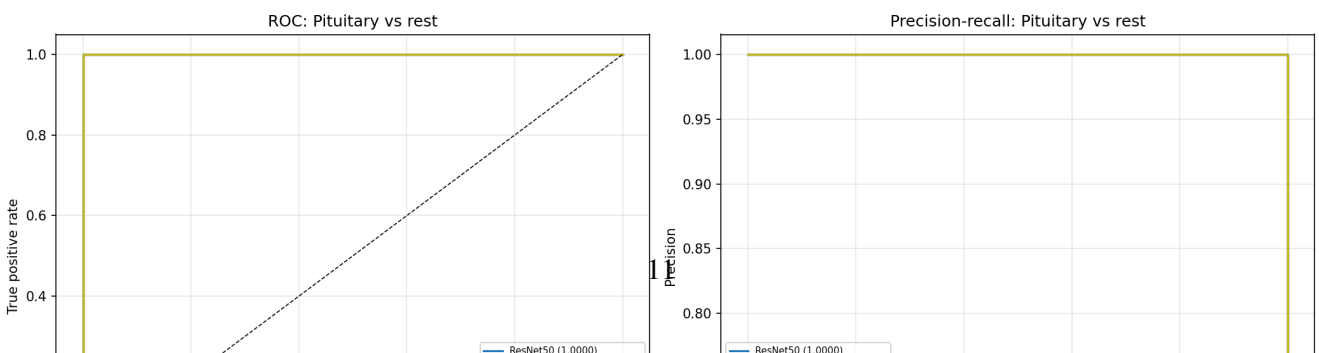
	Binary detection			Subtype classification	
Model	9–12	13–16	17–20	9–12	13–16
CoAtNet-0	0.9886	1.0000	1.0000	1.0000	1.0000
DenseNet121	0.9773	0.9589	1.0000	1.0000	1.0000
EfficientNetB0	1.0000	0.9863	1.0000	0.9773	1.0000
InceptionV3	1.0000	0.9589	1.0000	1.0000	1.0000
MobileNetV3-Large	0.9545	0.9863	1.0000	0.9773	1.0000
ResNet50	0.9886	0.9726	1.0000	1.0000	1.0000
Swin-Base	1.0000	0.9726	1.0000	1.0000	1.0000
VGG16	0.9659	0.9726	1.0000	0.9773	1.0000
ViT-B/16	1.0000	0.9863	1.0000	1.0000	1.0000

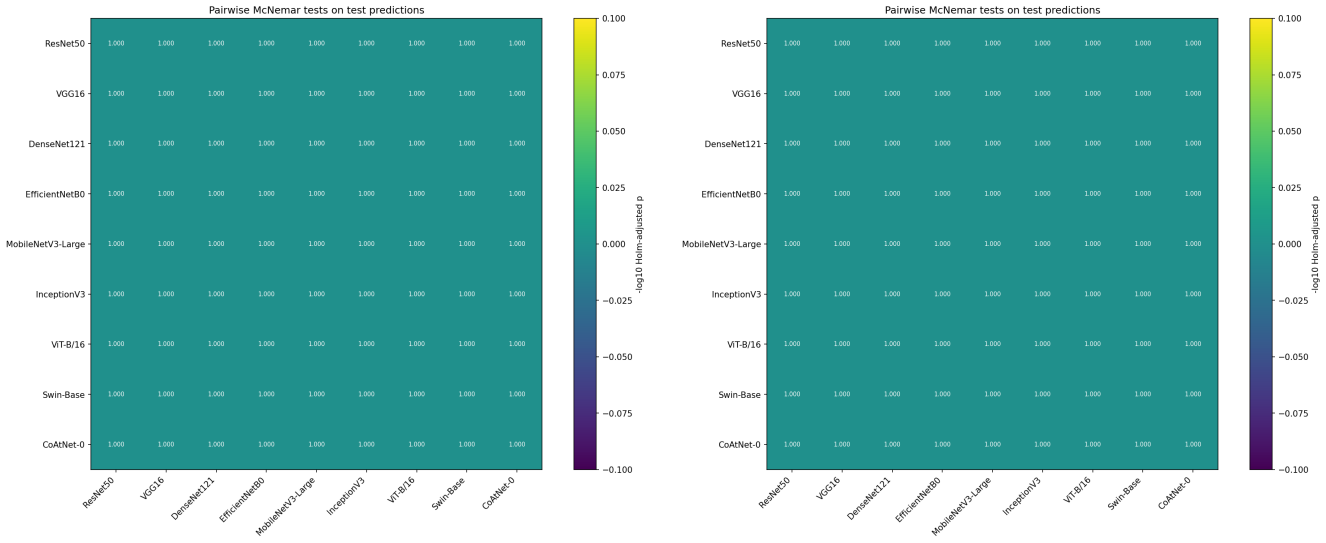


(a) Binary detection, tumour class.









(a) Binary detection.

(b) Subtype classification.

Figure S9: Holm-corrected exact McNemar p-values between the fold ensembles. Every adjusted p-value is 1.0 in both tasks (`mcnemar_holm_p.csv`).

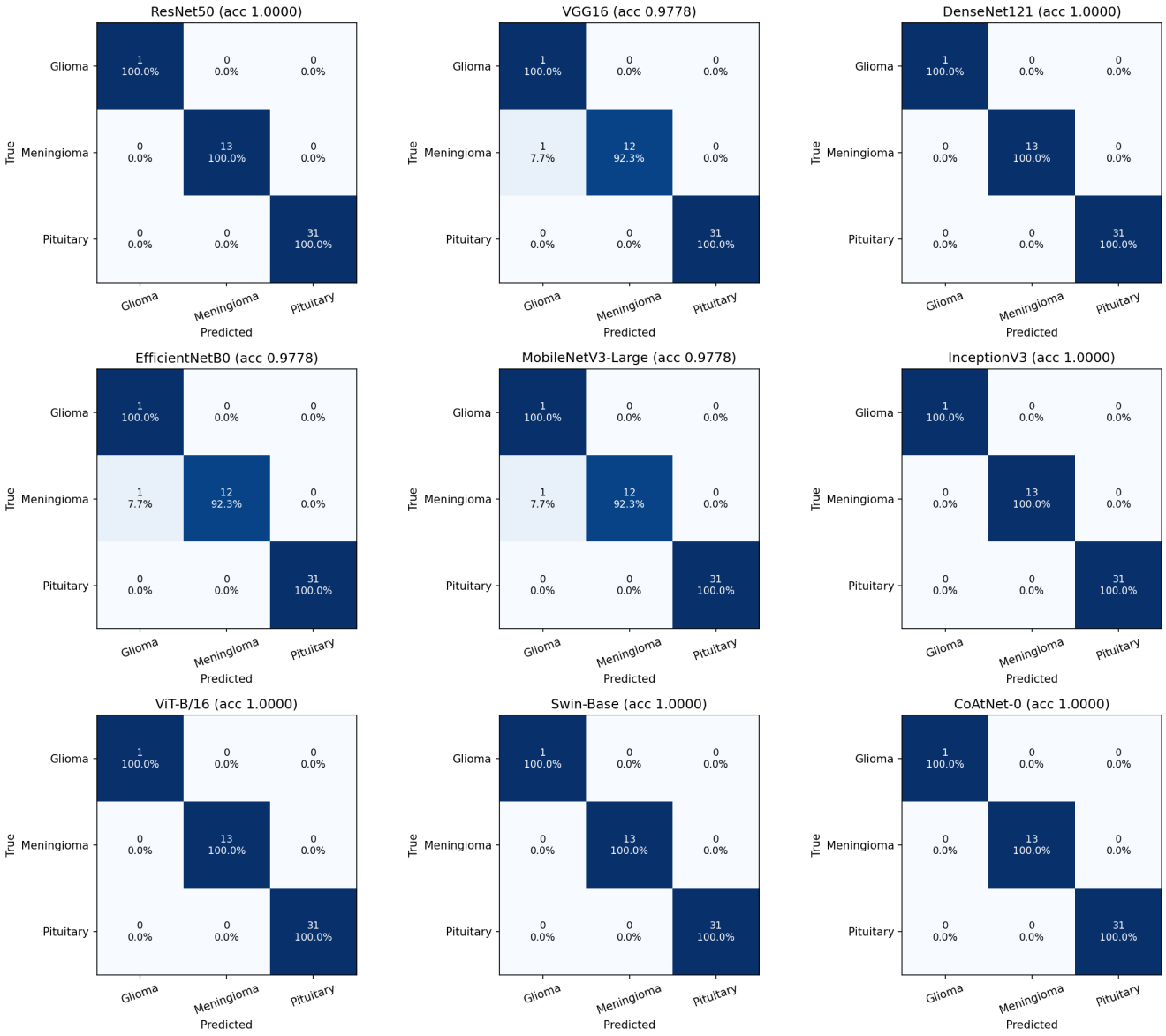
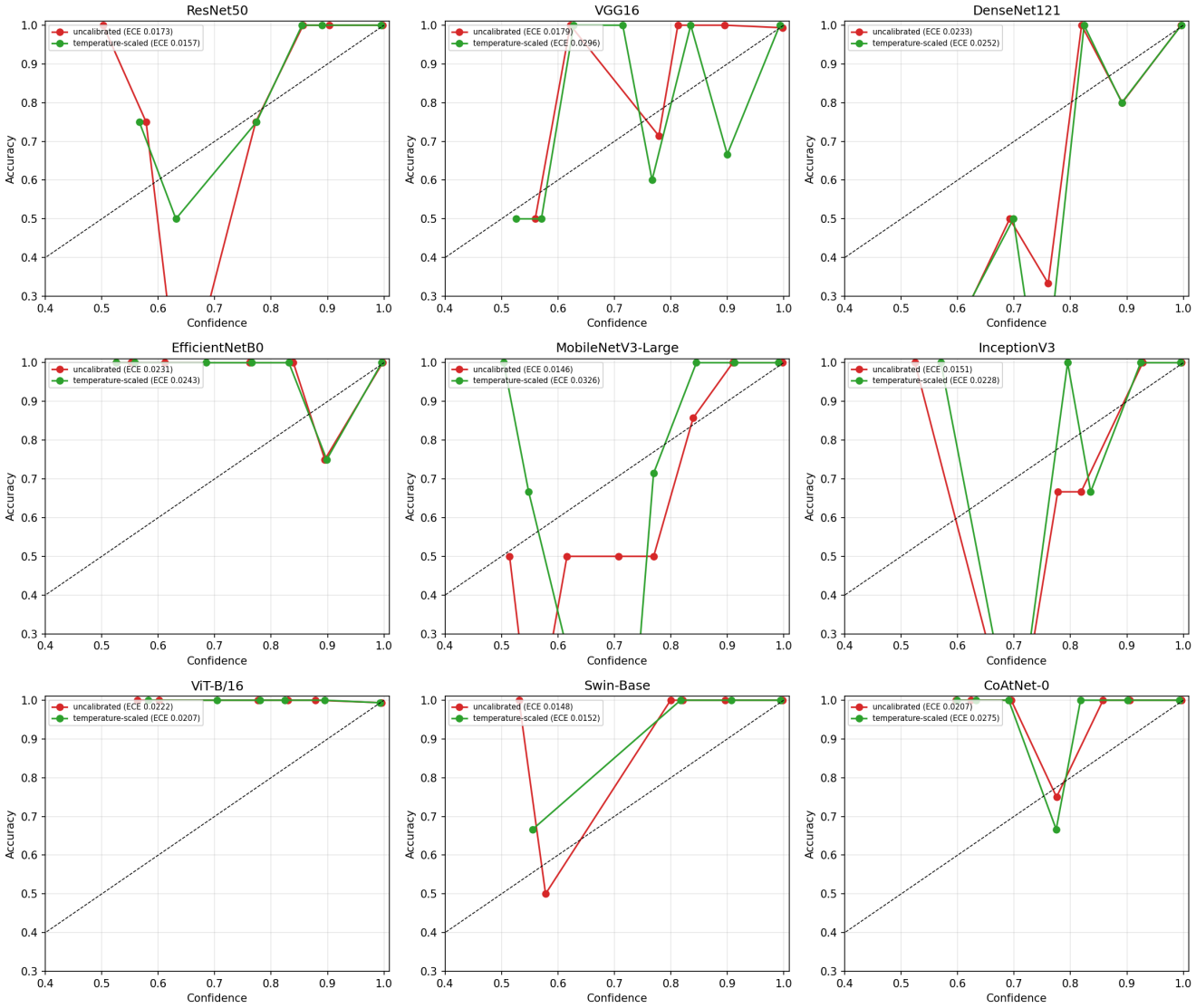
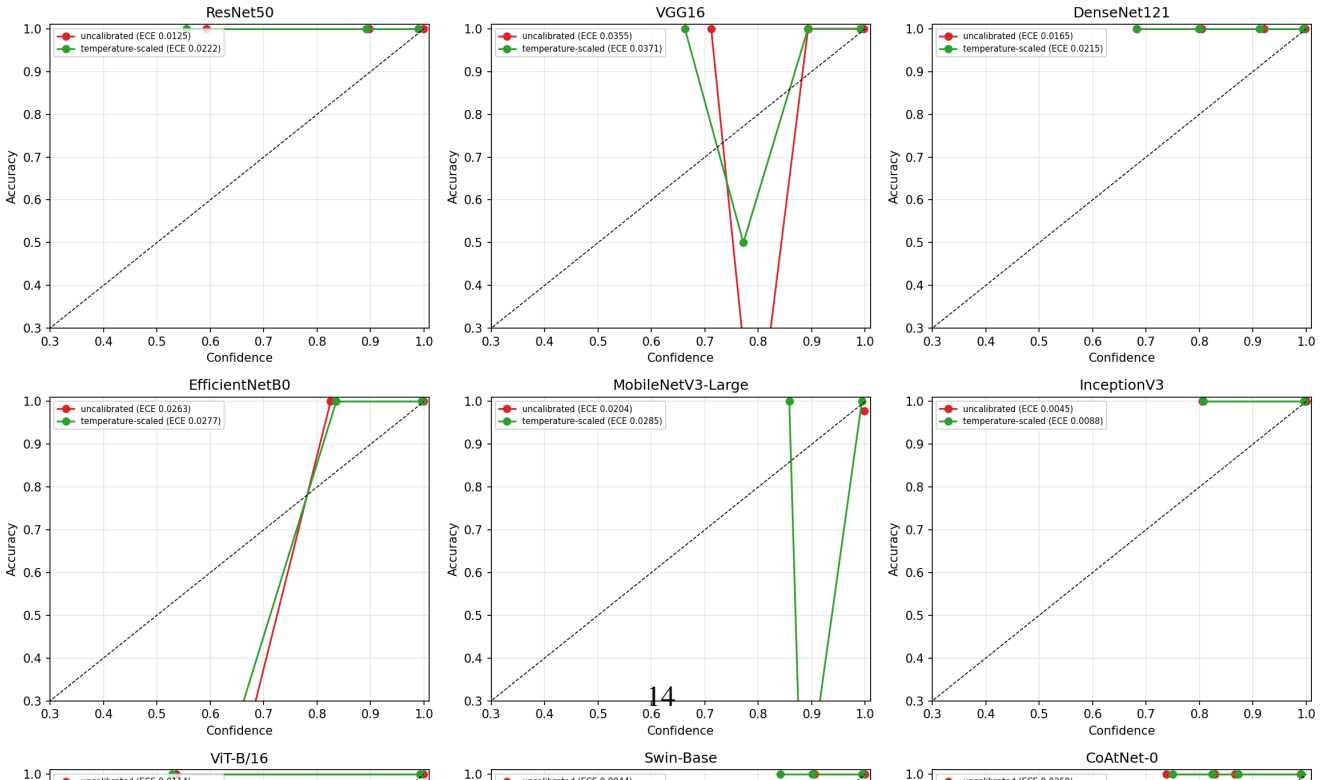
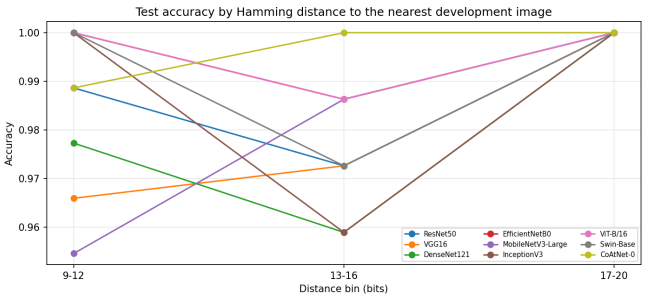


Figure S10: Subtype classification: test confusion matrices of every fold ensemble, counts with row percentages.

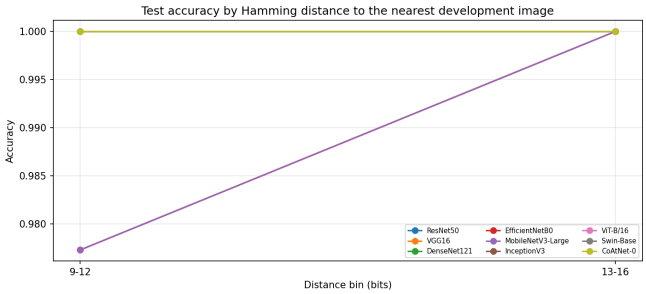


(a) Binary detection.





(a) Binary detection.



(b) Subtype classification.

Figure S12: Test accuracy by the Hamming distance from each test image to its nearest development image.

S4. Efficiency

Table S6: Complete efficiency measurements (`efficiency.csv`): latency (ms) and throughput at batch size 32 (images/s) as mean ± pooled standard deviation over three repeats on one Tesla T4 under half-precision autocasting; training throughput (images/s) and peak training memory (GB) averaged over the folds. Training throughput is a lower bound because the two workers shared four CPU cores.

Task	Model	Latency (ms)	Throughput (img/s)	Training (img/s)	Peak memory (GB)
Binary	ResNet50	9.57 ± 0.38	777.2 ± 6.1	235.2	5.62
	VGG16	7.72 ± 0.42	386.5 ± 4.8	124.4	9.88
	DenseNet121	23.58 ± 1.53	656.6 ± 22.9	185.4	4.48
	EfficientNetB0	13.13 ± 0.55	1393.3 ± 115.8	136.4	3.09
	MobileNetV3-Large	10.55 ± 1.09	2301.1 ± 292.9	182.3	2.09
	InceptionV3	17.63 ± 0.79	511.6 ± 5.8	155.8	5.85
	ViT-B/16	10.95 ± 1.04	272.2 ± 2.8	93.0	5.96
	Swin-Base	36.29 ± 1.23	197.6 ± 2.9	62.9	6.45
	CoAtNet-0	16.53 ± 1.07	513.1 ± 6.9	118.7	11.39
Multiclass	ResNet50	9.07 ± 0.36	716.8 ± 6.9	232.9	4.08
	VGG16	8.02 ± 0.40	332.1 ± 6.2	130.5	9.87
	DenseNet121	22.81 ± 1.22	607.7 ± 17.7	180.4	4.48
	EfficientNetB0	12.81 ± 0.80	1265.7 ± 50.3	243.9	3.04
	MobileNetV3-Large	9.99 ± 0.92	2183.1 ± 234.4	355.2	2.08
	InceptionV3	16.69 ± 1.25	438.2 ± 6.6	155.4	5.26
	ViT-B/16	10.26 ± 1.10	241.0 ± 3.0	82.6	5.97
	Swin-Base	34.65 ± 1.51	173.1 ± 2.8	57.8	6.45
	CoAtNet-0	15.88 ± 0.73	448.4 ± 6.3	131.1	10.42

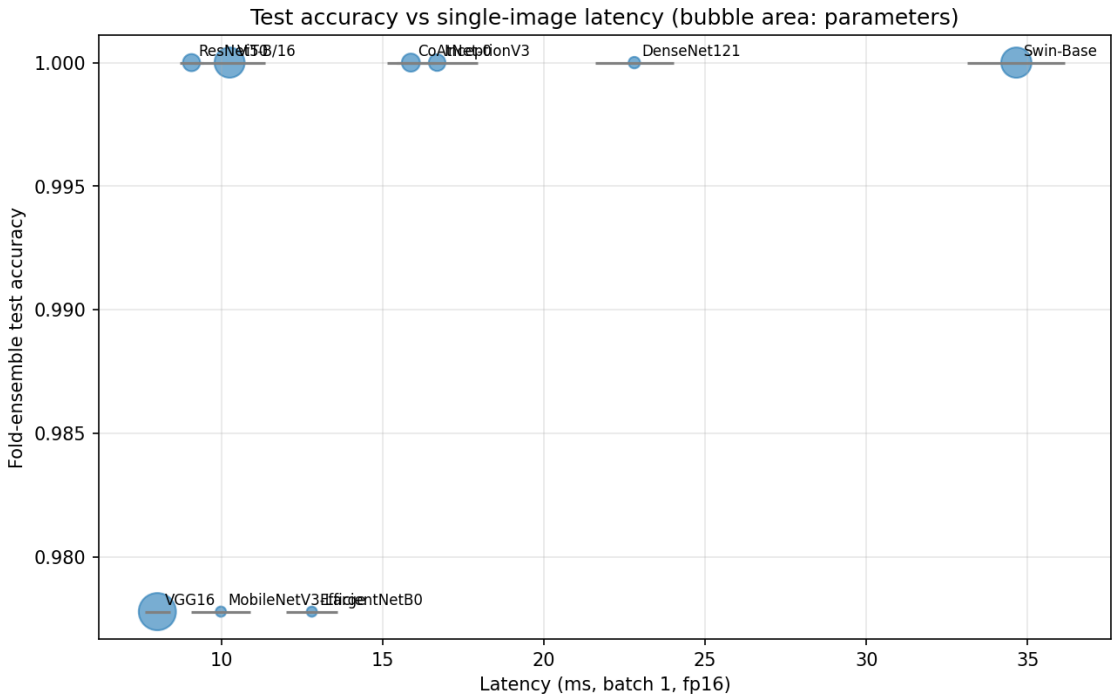


Figure S13: Subtype classification: fold-ensemble test accuracy against single-image latency for every model.

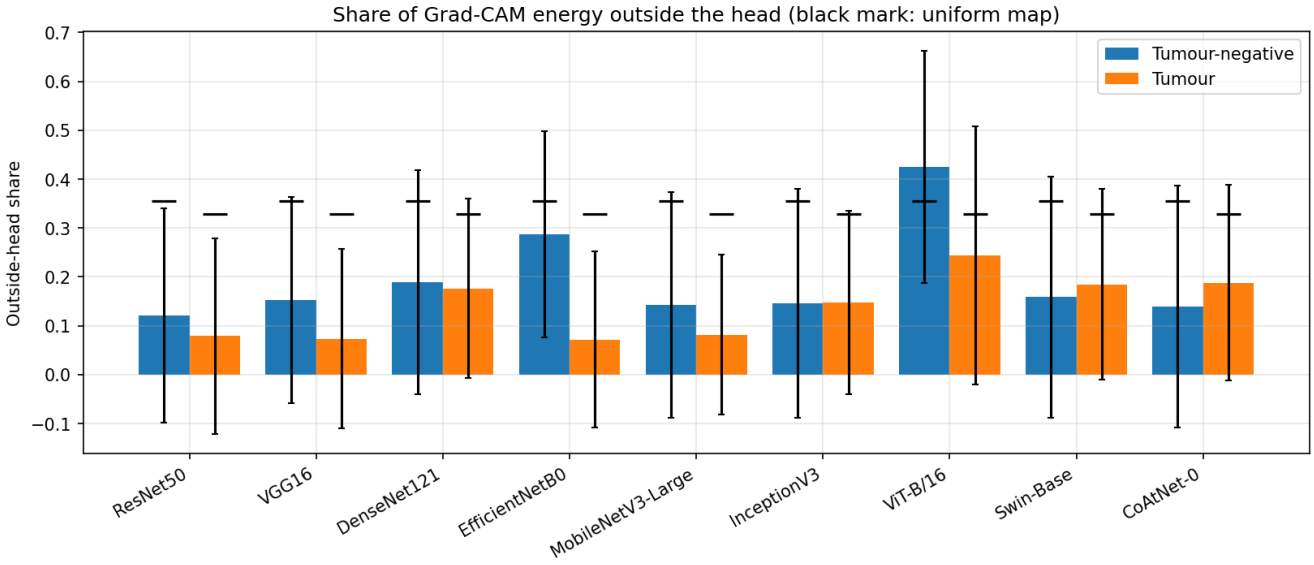
S5. Attention and shortcut analysis

Table S7: Share of Grad-CAM energy outside the head mask, mean $\pm$ standard deviation over test images (`outside_head_energy.csv`), and of occlusion sensitivity over 16 binary and 17 multiclass test images (`occlusion_outside_head.csv`). The multiclass test set has one glioma image, so that column has no standard deviation.

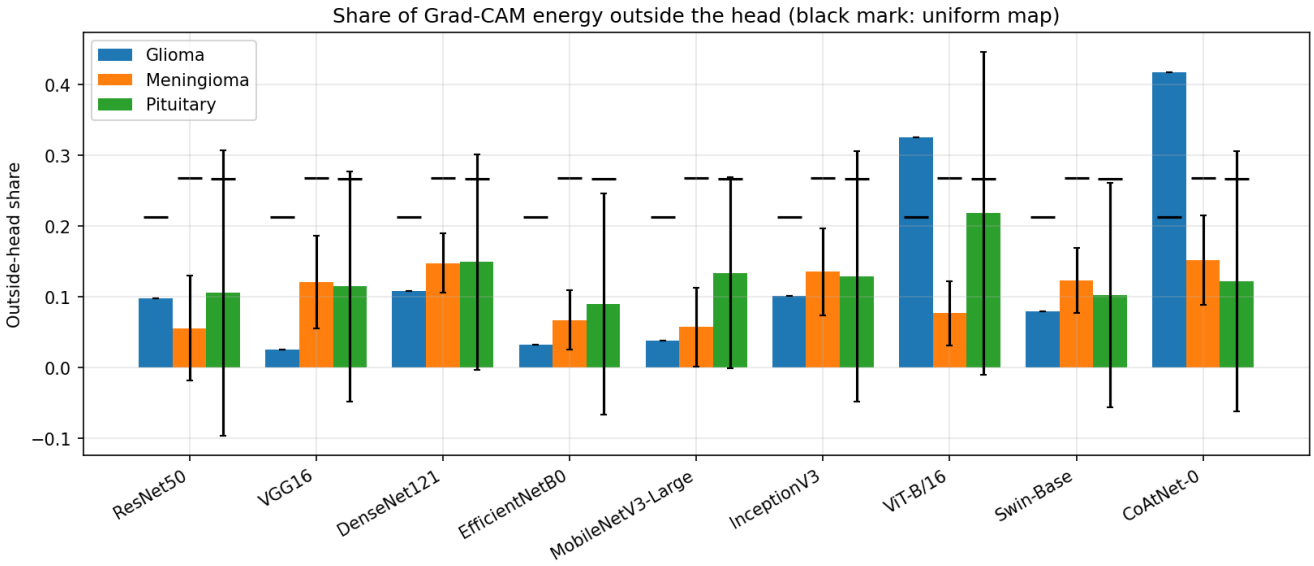
	Binary detection			Subtype classification			
Model	Tumour-negative	Tumour	Occlusion	Glioma	Meningioma	Pituitary	Occlusion
ResNet50	0.122 ± 0.219	0.079 ± 0.200	0.116	0.098	0.056 ± 0.074	0.106 ± 0.202	0.079
VGG16	0.153 ± 0.212	0.074 ± 0.184	0.007	0.026	0.121 ± 0.066	0.115 ± 0.163	0.079
DenseNet121	0.190 ± 0.229	0.176 ± 0.183	0.135	0.108	0.148 ± 0.042	0.150 ± 0.153	0.110
EfficientNetB0	0.287 ± 0.211	0.072 ± 0.180	0.194	0.033	0.067 ± 0.042	0.091 ± 0.156	0.066
MobileNetV3-Large	0.142 ± 0.231	0.082 ± 0.163	0.111	0.038	0.058 ± 0.056	0.134 ± 0.136	0.071
InceptionV3	0.146 ± 0.234	0.147 ± 0.187	0.167	0.102	0.136 ± 0.062	0.129 ± 0.178	0.099
ViT-B/16 (Grad-CAM)	0.426 ± 0.238	0.244 ± 0.264	0.155	0.326	0.077 ± 0.045	0.218 ± 0.229	0.107
ViT-B/16 (rollout)	0.421 ± 0.179	0.285 ± 0.163		0.262	0.350 ± 0.047	0.358 ± 0.121	
Swin-Base	0.159 ± 0.246	0.185 ± 0.195	0.182	0.079	0.123 ± 0.046	0.103 ± 0.158	0.122
CoAtNet-0	0.139 ± 0.248	0.188 ± 0.200	0.089	0.417	0.152 ± 0.064	0.122 ± 0.184	0.124
Uniform map	0.356	0.328		0.213	0.268	0.267	

Table S8: Shortcut test (`shortcut_test.csv`): models trained (fold 0) and tested on background-only images. The p-value is a one-sided exact binomial test against the majority-class rate (0.6257 binary, 0.6889 multiclass), Holm-corrected within each task.

	Binary detection			Subtype classification		
Model	Accuracy	AUC	Holm p	Accuracy	AUC	Holm p
ResNet50	0.8129	0.8657	$< 10^{-4}$	0.9333	0.9968	0.0006
VGG16	0.7485	0.8359	0.0013	0.9778	1.0000	$< 10^{-4}$
DenseNet121	0.7953	0.8578	$< 10^{-4}$	0.8667	0.9904	0.0104
EfficientNetB0	0.7778	0.8584	0.0001	0.8889	1.0000	0.0048
MobileNetV3-Large	0.6784	0.7821	0.1772	0.9333	1.0000	0.0006
InceptionV3	0.7953	0.8606	$< 10^{-4}$	0.9778	0.9984	$< 10^{-4}$
ViT-B/16	0.6550	0.7411	0.2396	0.9111	0.9717	0.0020
Swin-Base	0.8304	0.8893	$< 10^{-4}$	0.9111	0.9976	0.0020
CoAtNet-0	0.7953	0.8756	$< 10^{-4}$	0.8222	0.9752	0.0337

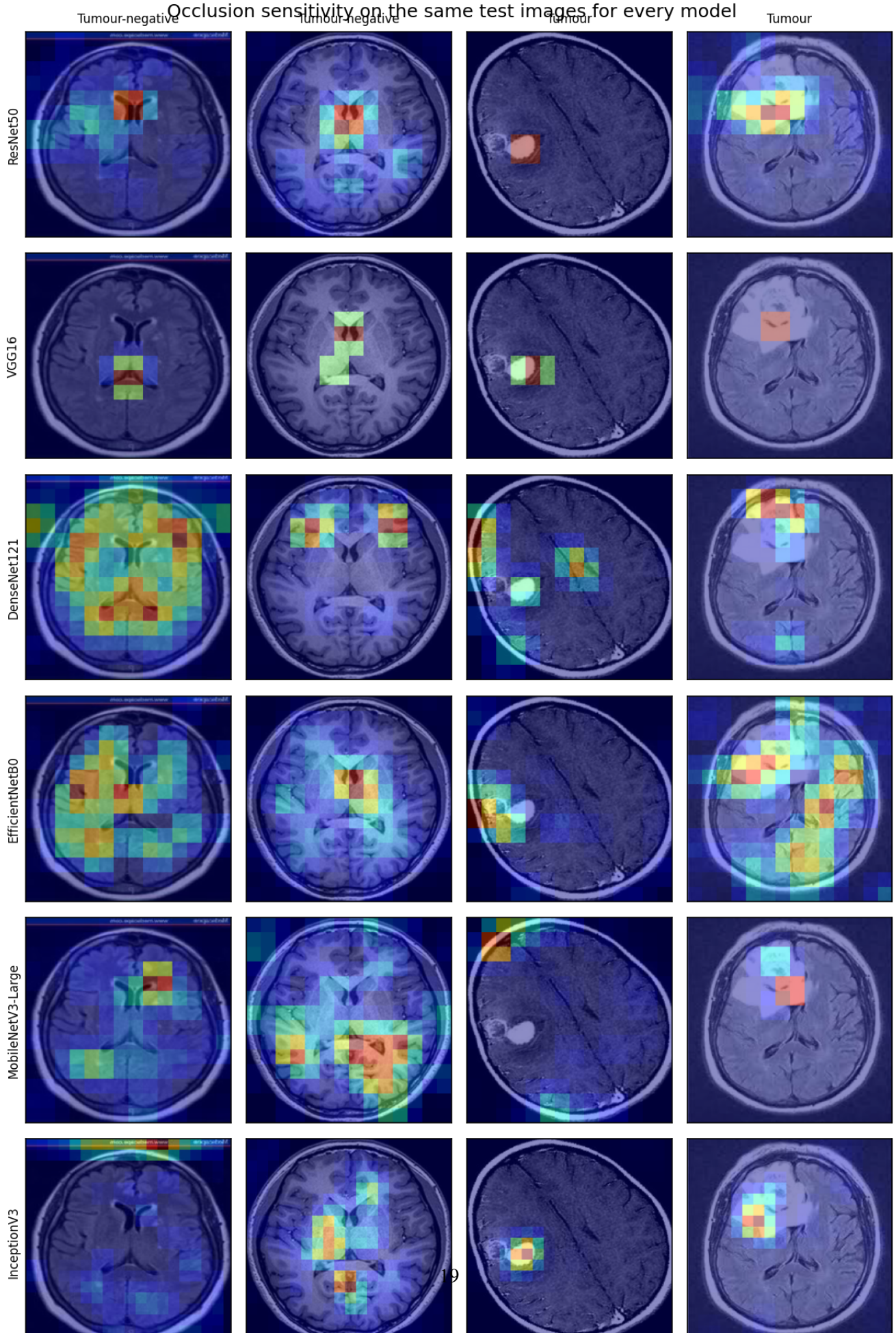


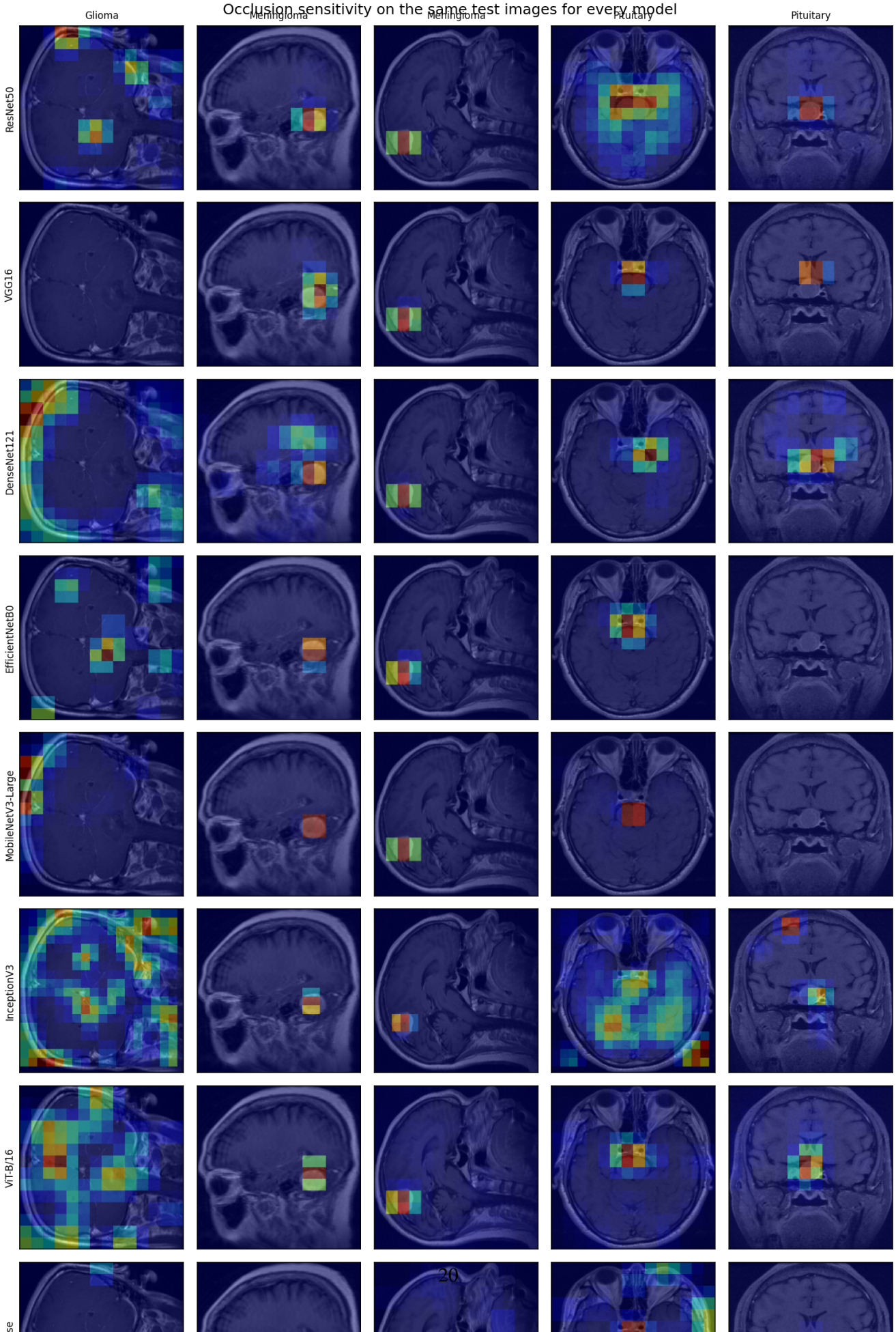
(a) Binary detection.

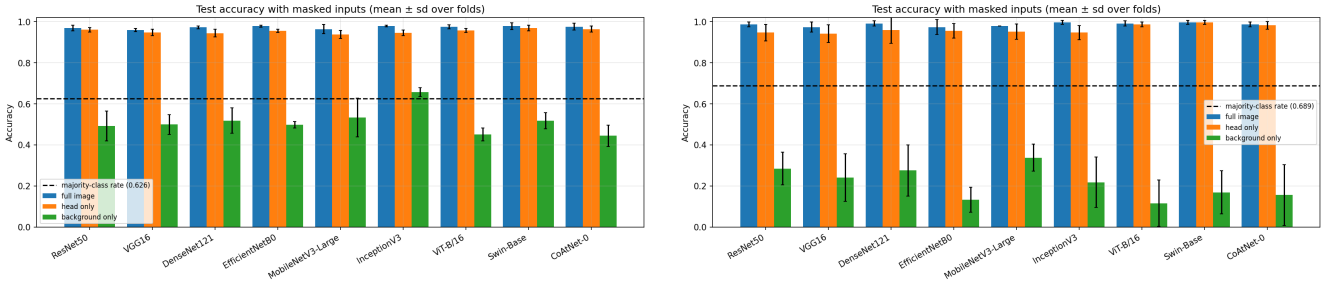


(b) Subtype classification.

Figure S14: Share of Grad-CAM energy outside the head per model and class; the black mark is the share of a uniform map.


Occlusion sensitivity on the same test images for every model
Tumour-negative
Tumour-negative
Tumour
Tumour
ResNet50
VGG16
DenseNet121
EfficientNetB0
MobileNetV3-Large
InceptionV3


Occlusion sensitivity on the same test images for every model
Glioma
Meningioma
Meningioma
Pituitary
Pituitary
ResNet50
VGG16
DenseNet121
EfficientNetB0
MobileNetV3-Large
InceptionV3
ViT-B/16

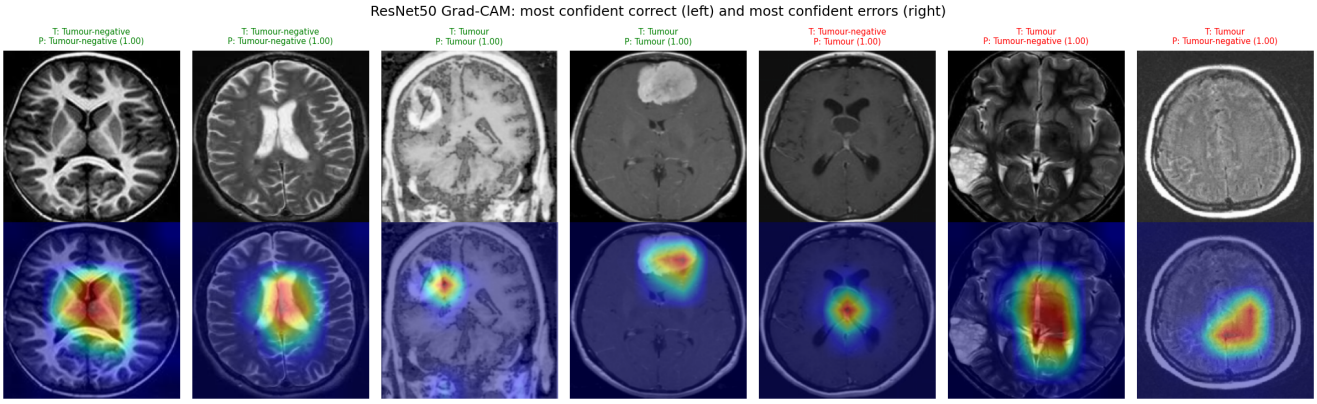


(a) Binary detection.

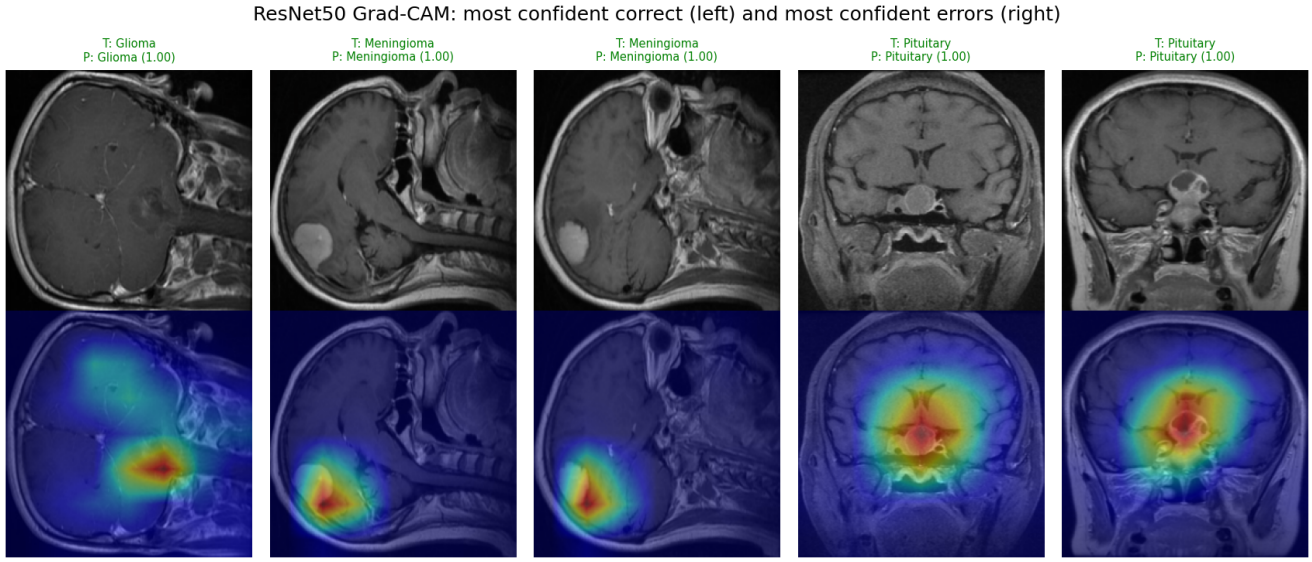
(b) Subtype classification.

Figure S17: Masking test (inference only): test accuracy of each fold's checkpoint on full images, head-only images (background set to zero) and background-only images (head set to zero), mean ± standard deviation over folds.

S6. Grad-CAM maps of every model

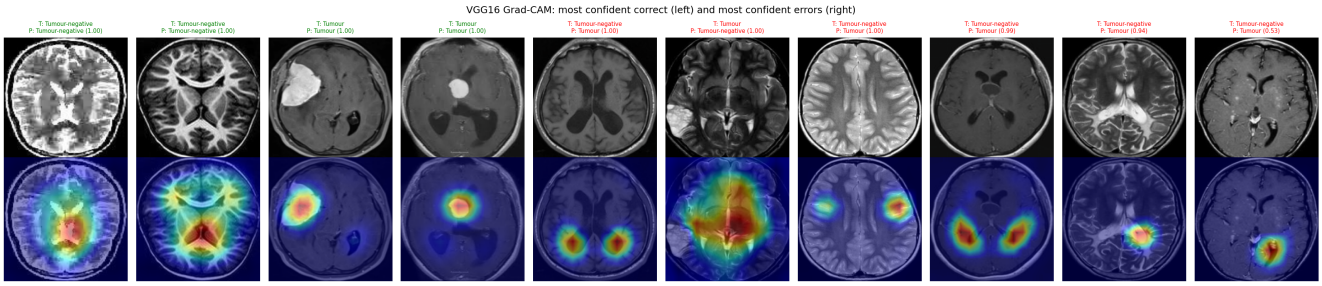


(a) Binary detection.

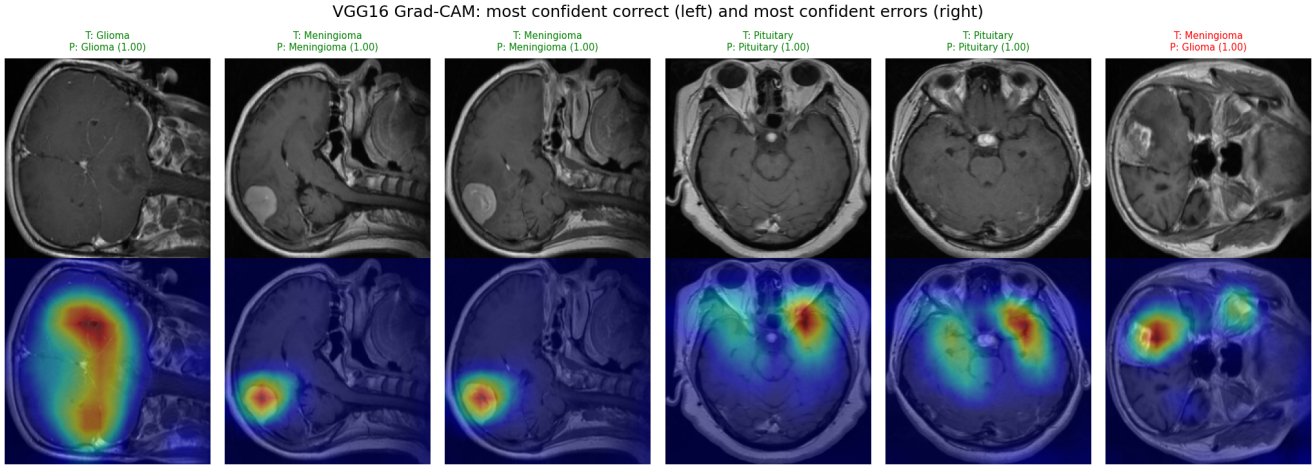


(b) Subtype classification.

Figure S18: Grad-CAM maps of **ResNet50** on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

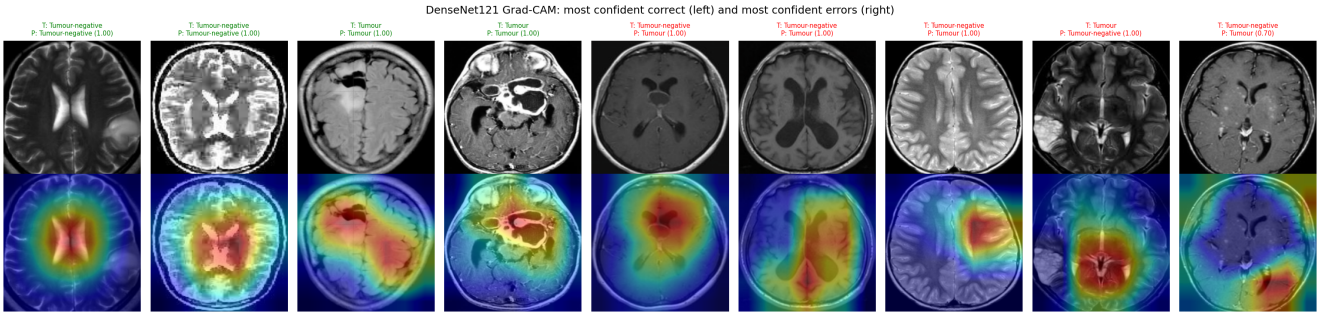


(a) Binary detection.

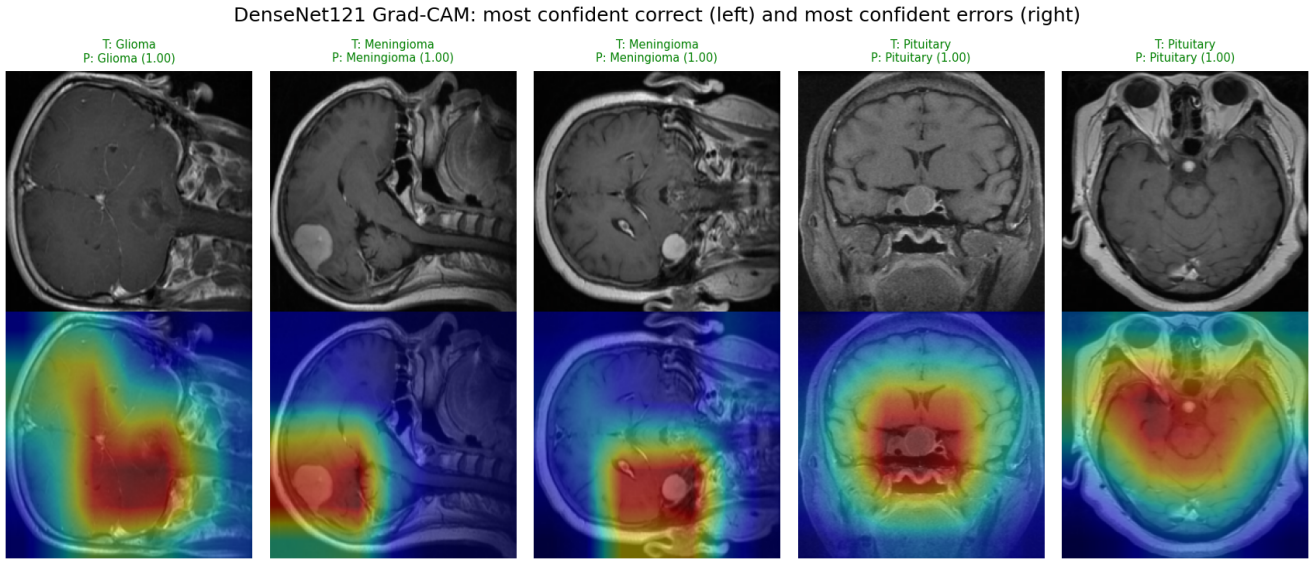


(b) Subtype classification.

Figure S19: Grad-CAM maps of **VGG16** on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

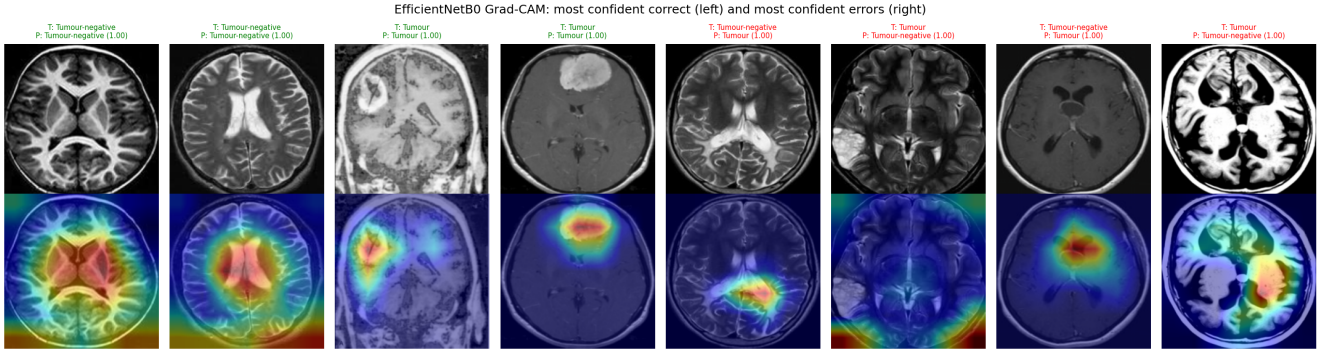


(a) Binary detection.

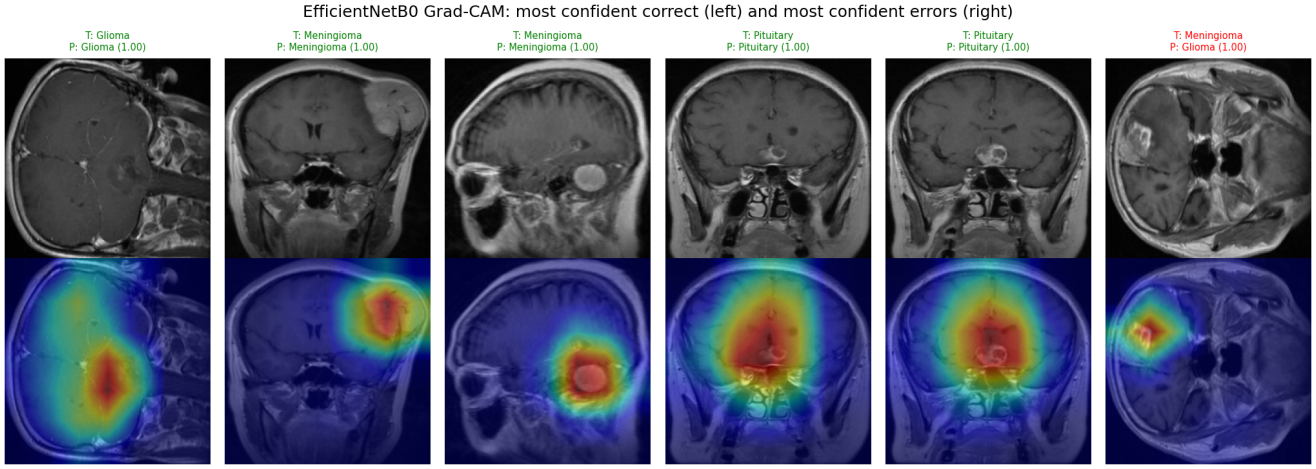


(b) Subtype classification.

Figure S20: Grad-CAM maps of **DenseNet121** on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

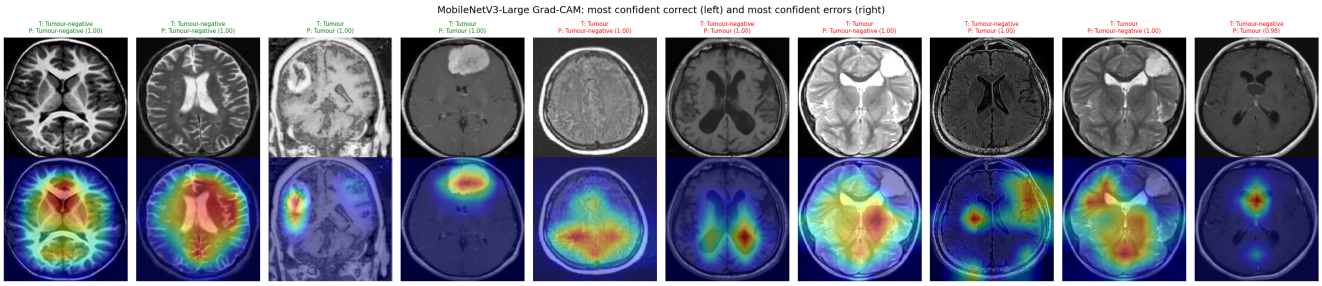


(a) Binary detection.

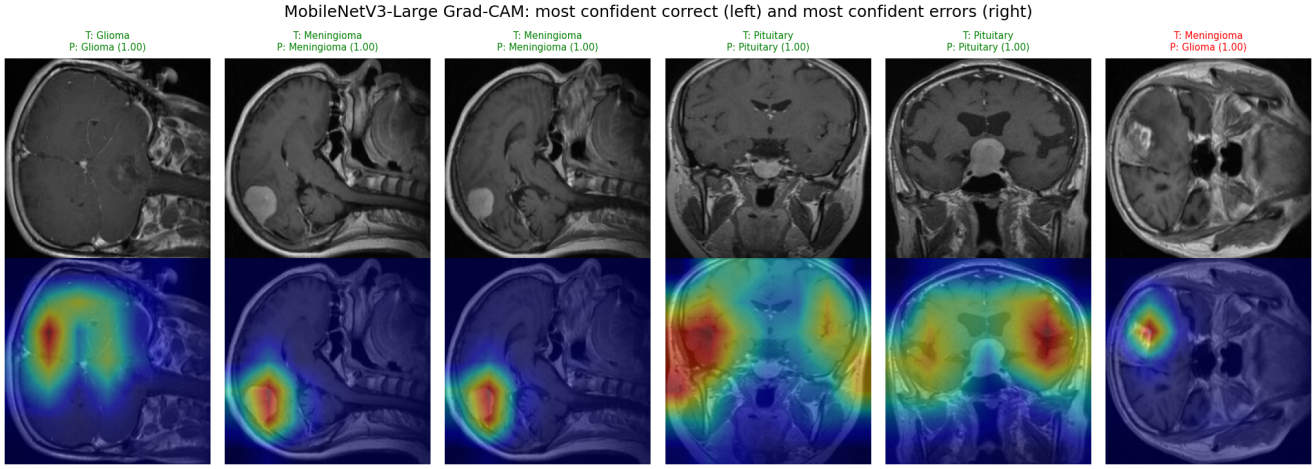


(b) Subtype classification.

Figure S21: Grad-CAM maps of **EfficientNetB0** on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

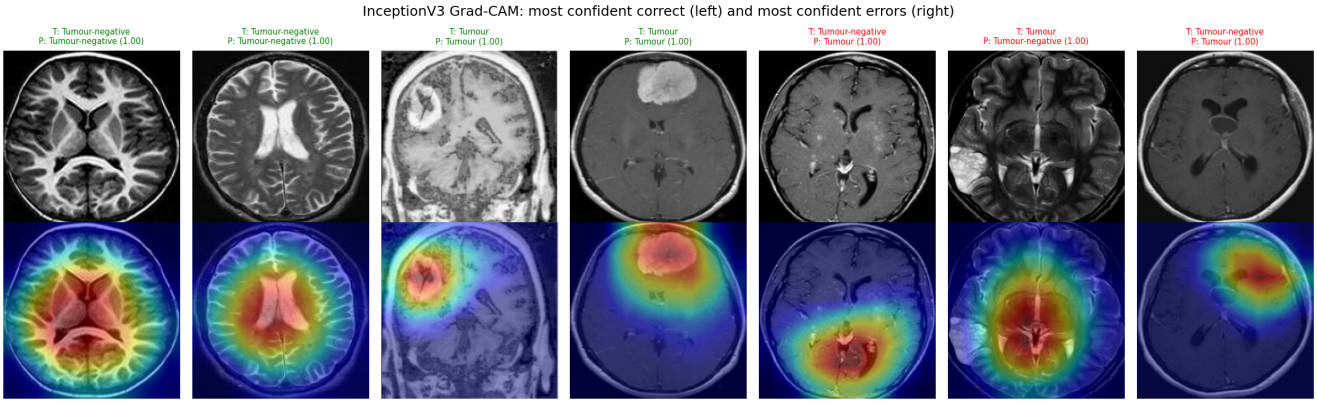


(a) Binary detection.

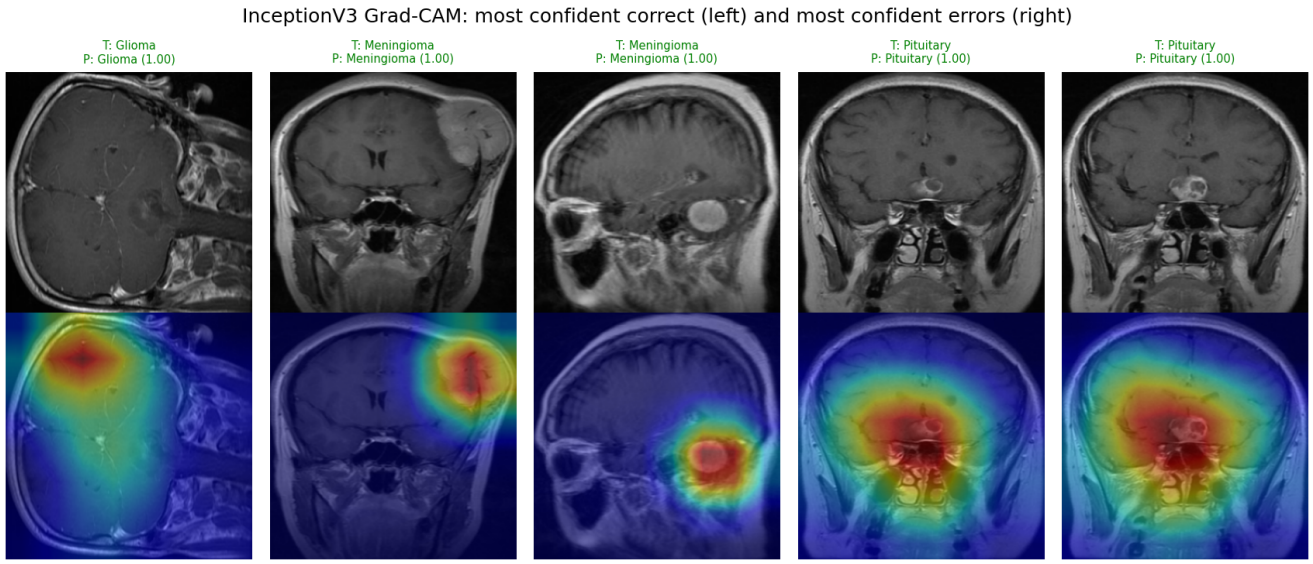


(b) Subtype classification.

Figure S22: Grad-CAM maps of **MobileNetV3-Large** on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

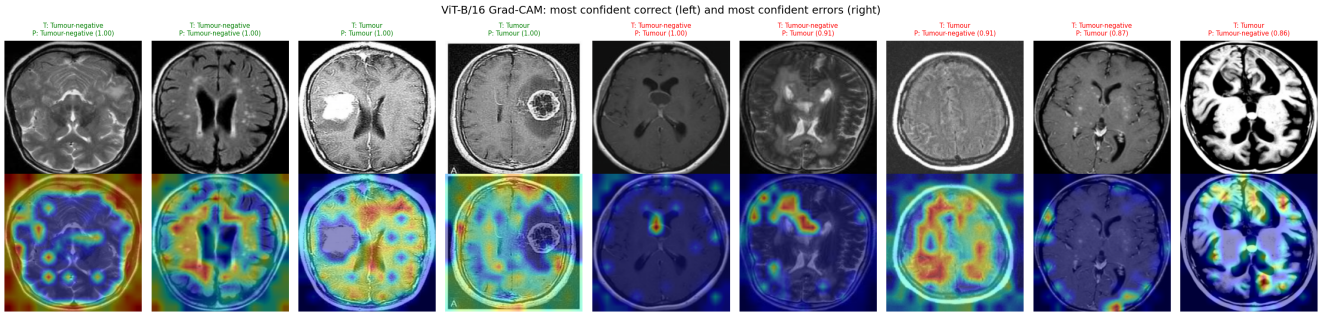


(a) Binary detection.

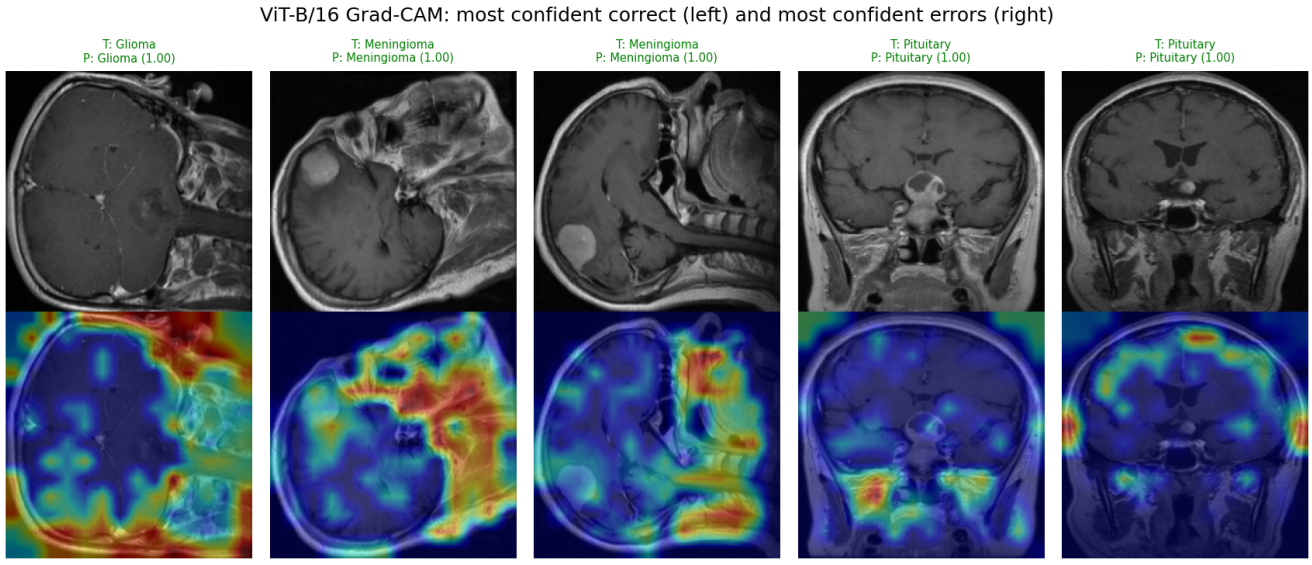


(b) Subtype classification.

Figure S23: Grad-CAM maps of **InceptionV3** on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

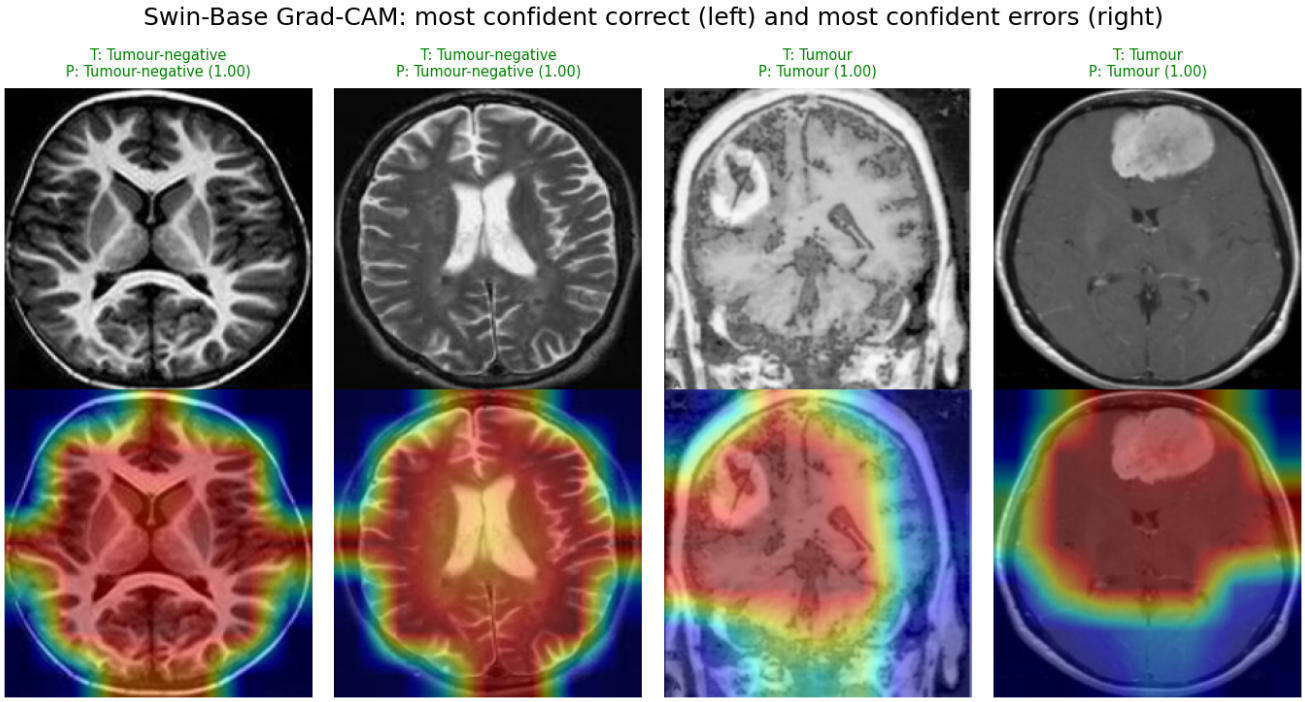


(a) Binary detection.

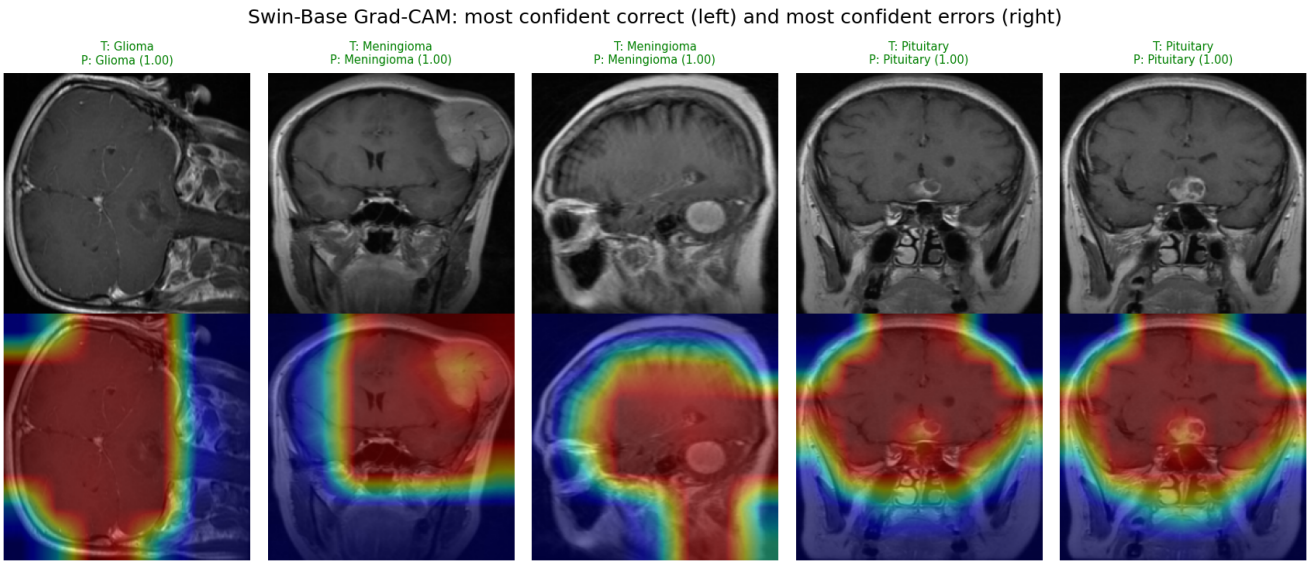


(b) Subtype classification.

Figure S24: Grad-CAM maps of **ViT-B/16** on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

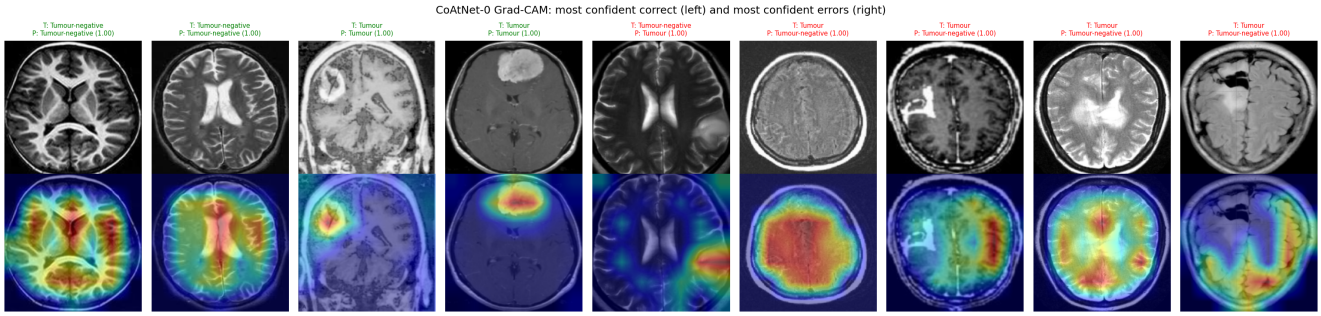


(a) Binary detection.

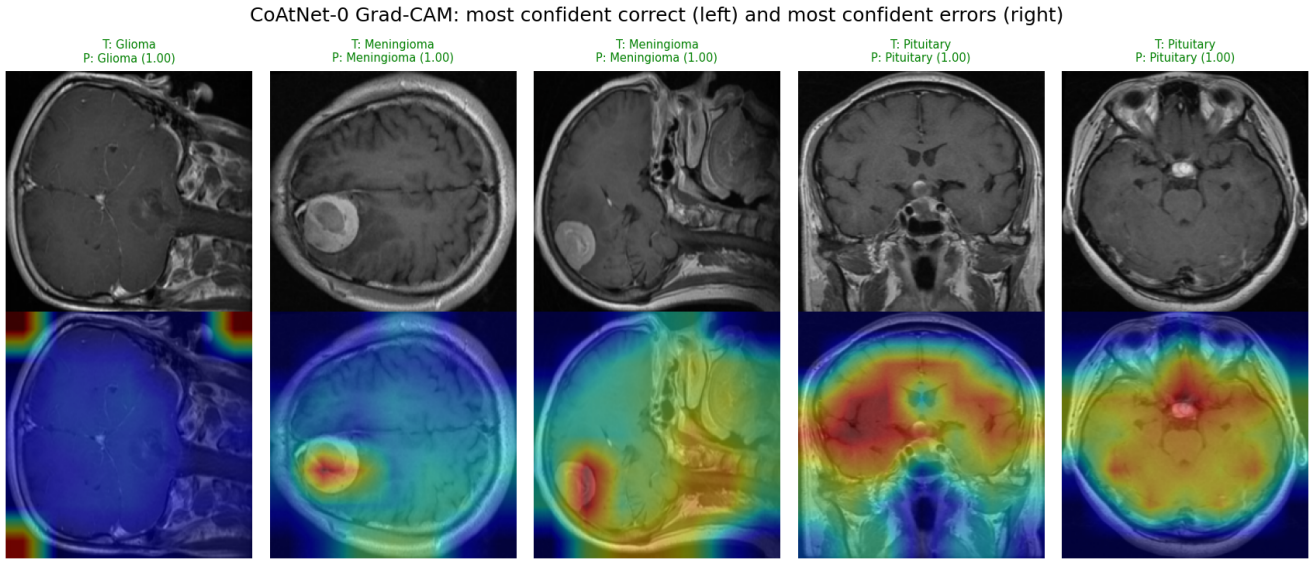


(b) Subtype classification.

Figure S25: Grad-CAM maps of **Swin-Base** on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.



(a) Binary detection.



(b) Subtype classification.

Figure S26: Grad-CAM maps of **CoAtNet-0** on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.